Bachelor Degree Project

Co-design, implementation and evaluation of a teacher-facing Learning Analytics Dashboard prototype for Nextgen Classroom.



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Abstract

This thesis explores the co-design, implementation, and evaluation of a Learning Analytics (LA) dashboard prototype for the Nextgen classroom web application, aiming to enhance data-informed learning and teaching practices in K-12. Although digital classroom platforms generate large volumes of learner data, these data are often underutilized or presented without pedagogical grounding, limiting their capacity to support self-regulated learning, engagement monitoring, and timely instructional intervention. To address this gap, the authors co-designed and developed a self-contained, theoretically informed analytics dashboard prototype using React and TypeScript. This prototype transforms raw activity and performance data into meaningful visualizations aligned with self-regulated learning constructs. Because of Nextgen Classroom's strict security policies preventing external modifications to their production branch, the final integration of the dashboard into the main application was carried out internally by the company's team. The research follows a structured methodology consisting of Design-Based Research (DBR), Co-Design and Participatory Design and a mixed-methods evaluation strategy combining usability testing, quantitative and qualitative user feedback. Results indicate that usability and perceived usefulness was above the midpoint of the scale. These results are based on a small sample size (n = 10) and showed no statistically significant changes, limiting the strength and generalizability of the conclusions. The qualitative findings, however, identified several requirements that teachers considered important for adopting this dashboard.

Keywords: *Nextgen Classroom, Dashboard, Learning Analytics, statistics, AI in education, educational technology, teaching*

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Author disclaimer

This thesis has incorporated the use of artificial intelligence (AI) tools to assist in the rewriting and refinement of certain sections of text, as well as in the retrieval and verification of references and supporting information. All AI-assisted contributions were critically reviewed, edited, and validated by the author to ensure academic integrity, accuracy, and adherence to the research objectives. The final content reflects the author's independent analysis, interpretation, and scholarly judgment.

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1 Introduction

Recognizing students who are struggling or falling behind is a time-consuming challenge teachers have to navigate [1]. Without tools to provide insight into students' individual progress it can become overwhelming for teachers [2]. As a consequence, students who fall behind risk not receiving support when it is needed most.

Educational technology (EdTech) has advanced significantly over the past two decades [3]. The advancements have resulted in increased attention, engagement, motivation and performance of students, particularly when it is used to support student effort rather than for content delivery [3], [4]. For teachers, the advancements have provided better education tools to support students [5]. Digital learning platforms are one such tool that assist teachers in providing insight into student progress [6].

Nextgen Classroom, a web application for language learning is one such platform and a very recognised one which has won national and regional awards in Sweden [7], [8], [9]. This platform collects large volumes of learner data generated through student activity [10]. This data is often underutilized or presented without pedagogical grounding [6], [11]. This limits its capacity to support self-regulated learning, engagement monitoring, and timely instructional intervention [12].

This thesis addresses this challenge by focusing on co-designing, implementing and evaluating a theoretically informed Learning Analytics (LA) dashboard prototype into the Nextgen Classroom web application as work done during an internship in the company. This platform transforms students' raw activity and performance data into meaningful visualizations. The goal is to enhance data-informed learning and teaching, and supporting reflective decision-making in K-12 educational settings.

1.1 Background

LA is a young field that has rapidly taken shape over the last decade. Although publications started as early as 2010 [13], it did not become an established academic discipline until 2011 at the first International Conference on Learning Analytics and Knowledge [14]. This research area is commonly described as “the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environment in which it occurs.” [15] and in digital classroom platforms large volumes of raw learner data is generated through student engagement and performance. LA dashboards are used to transform learner data into meaningful visualizations and provide data-driven insight for teachers to support their students effectively [14], [15], [16].

Descriptive analytics and static visualization are prevalent in existing LA dashboards [16], [17], making it difficult to efficiently support teachers and requiring teachers to manually analyze trends and offering limited interaction and support for sensemaking and reflective decision-making. Although learner data is visualized through LA dashboards, one of the challenges is that most descriptive dashboards employ surface-level analytics [18] and provide limited actionable insights [19] and support for teacher sensemaking and decision-making.

1.2 Related work

1.2.1 Teacher perception of educational technology

Teachers' attitude towards and willingness to use EdTech are paramount to how EdTech is integrated into the classroom. Akram et al. [20] found that there is a positive attitude towards technology being incorporated in education. This aligns with findings from Abel et al. [21] which reported a positive attitude towards the integration of EdTech and that it is believed to enhance the performance of students. However, a positive attitude is not a guarantee of good use. Hase et al. [6] found that teachers struggle analyzing and interpreting data as well as making data-based decisions. Michos et al. [22] findings reflect this as despite teachers having access to technology, only one third made use of these technologies, and only a quarter felt confident in using it to teach.

LA emerged as a way to help teachers make sense of data. The study by Valtonen et al. [23] revealed that teachers found this useful for understanding class learning activities and identifying students who are in need of additional support.

1.2.2 Challenges of Language learning with learning analytics

Although teachers perceive usefulness in LA, Kew et al. [24] found that behavioral data is mainly the focus in LA. This is reflected by Azad [12] and Mohseni et al. [1] who heavily rely on behavioral proxy indicators to represent student engagement. Gašević et al [25] argue that surface-level behavioral data such as login frequency does not reflect effective learning. In language learning, progress cannot be measured with behavior data alone, clicks and logins capture activity but do not represent linguistic growth. Colling et al. [26] tackle this challenge by co-designing a teacher-facing LA dashboard for language learning together with teachers, built on the Intelligent Language Tutoring System, FeedBook, that offers exercises to students. These exercises give feedback in the form of whether the student was correct or not and hints at mistakes without providing the correct answer. The exercise answers are used to form the baseline for all calculations in the dashboard, so teachers see language learning progress, rather than only using behavioral proxy data.

1.2.3 Limitation of visualization in LA dashboards

Limits are imposed not only on the data that existing LA dashboards collect, but also on how the data is presented. Findings from Vieira et al. [27] show that LA dashboards often rely on basic visualizations with limited interactivity and are not informed by pedagogical practices. Dollinger et al. [28] suggest that the limited collaboration between teachers and researchers is part of the reason why the design of LA dashboards does not reflect the pedagogical needs of teachers who use them. However, Nguyen et al. [29] together with teachers co-designed TEADASH, a teacher-facing LA dashboard which extracts data from Canvas LMS, to visualize learner activity. Although teachers were satisfied with the overall usefulness of the dashboard, some teachers did not know how to act on the information shown by its network graph or prediction table. This suggests that co-designing is not a guarantee of an actionable dashboard. .

1.2.4 Co-designing LA dashboards to support teachers

Even so, recent work has co-designed dashboards with teachers to align visualization to their needs with positive results. Mohseni et al. [30] co-designed EasyVis, teacher-facing multiple LA dashboards, together with teachers from three different municipalities in Sweden, aggregating data such as learning activities provided by EdTech companies, attendance and final grades to create visualizations to better support insights at both class and student levels. Karademir et al. [31] co-designed a teacher-facing LA dashboard together with teachers in K-12. They compared classes where teachers had access to the dashboard with classes where teachers did not. Student knowledge gain was measured with a pre- and post-test. The classes where teachers used the LA dashboard displayed significantly higher knowledge gains compared to those without access to it.

1.3 Problem formulation

Language learning is widely recognized as a complex and multifaceted process, which becomes even more challenging in instructional contexts where teachers must support diverse learners simultaneously. Effective language acquisition typically requires personalized pedagogical approaches tailored to individual learners' needs and progress [32]. There is a lot of learner diversity among students in the same class, depending on their background, learning goals and proficiency level. However, many existing educational dashboards fail to meaningfully support such personalization.

Current dashboard systems often provide separate views at the class and individual levels [29]. Nevertheless, these systems largely rely on behavioral proxy indicators (such as login frequency, click activity, or time spent on tasks) that do not accurately reflect learners' actual linguistic development [1], [12]. As a result, these dashboards offer limited actionable insights and restrict teachers' ability to make informed, data-driven pedagogical decisions [33].

To address this limitation, this thesis proposes a dashboard architecture designed to better represent language learning progression and will be evaluated empirically with real language teachers. The design requirements come from teacher focus groups and the proposed system integrates aggregated class-level analytics (capturing overall performance and engagement) with the capability to drill down, moving from class view into detailed individual-level data, including learner performance and progress indicators.

Based on the focus groups, the proposed approach bridges class-level and individual-level learning analytics and aims to support teachers in identifying both collective learning patterns and individual learner needs. This enables instructors to more effectively diagnose areas where students struggle and to adapt instruction accordingly. Consequently, teachers can design targeted classroom interventions while also providing additional individualized support when necessary, thereby facilitating more personalized and effective language learning experiences.

Building upon the challenges identified in the preceding sections and the practical insights gained during the internship at Nextgen Classroom, this study seeks to address a central pedagogical question: how can language teachers more effectively interpret classroom-level and individual learning needs through data-driven tools? To explore this issue, the following main research question and sub-questions were formulated.

Main Research Question

RQ How do teachers perceive the usefulness and usability of a proposed learning analytics dashboard designed for the Nextgen Classroom platform for monitoring language learning progress and supporting pedagogical decision-making?

Sub-questions

RQ1a – Usability and Ease of Use

How do teachers perceive the usability and ease of use of the proposed Learning Analytics dashboard prototype after hands-on interaction in workshop settings?

RQ1b – Interpretation of Visual Representations

How do using the dashboard influence teachers' interpretation and perceived usefulness of its visual representations of learning progress, compared to before use?

RQ1c – Adoption and Integration Factors

What factors related to design, data, and integration with existing tools affect teachers' acceptance of the dashboard for ongoing pedagogical decision-making?

1.4 Motivation

The increasing digitalization of education has led to the large-scale collection of student engagement and performance data, yet teachers often struggle to interpret this information efficiently and translate it into actionable pedagogical decisions. From a scientific perspective, this challenge is highly relevant to research in Learning Analytics, Human–Data Interaction, and EdTech, as the design of an interactive dashboard provides empirical insight into how data visualizations support decision-making and evidence-based teaching within real classroom contexts. From a societal perspective, enabling teachers to access clear, data-driven insights is essential for personalizing learning, identifying students who need timely support, and promoting more equitable and effective educational practices. From an industrial perspective, platforms such as Nextgen Classroom already gather rich datasets, but without appropriate analytical tools their practical value remains underexploited; an interactive learning analytics dashboard can summarize key metrics, track trends, and highlight students requiring attention, thereby enhancing the platform's functionality and value for its users.

1.5 Milestones

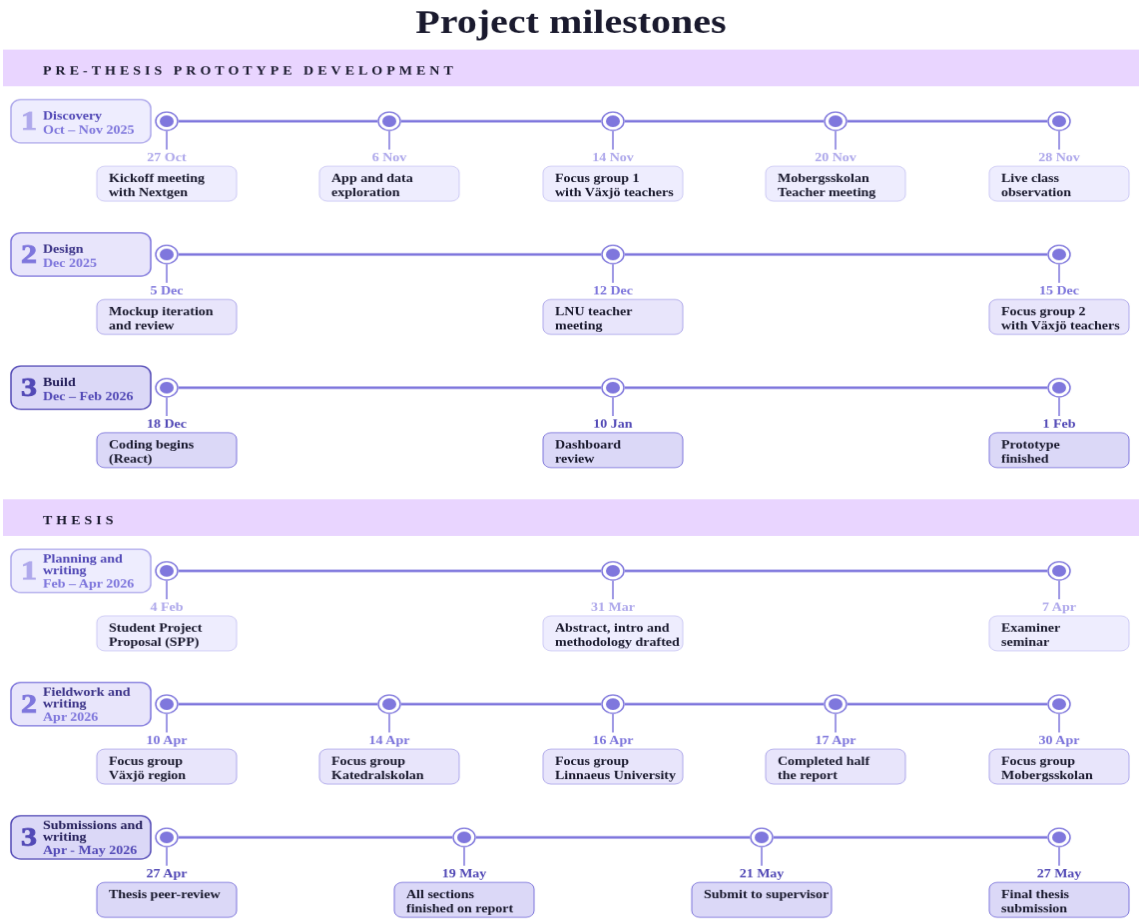


Figure 1.1: Timeline of project milestones for the dashboard project development.

1.6 Scope/Limitation

The study involved teachers and IT-pedagogues, with a total of ten participants through different focus groups and workshops, limiting statistical significance and generalizability. Nonetheless, it achieved its objective of exploring teachers' perceptions and practical needs regarding language Learning Analytics dashboards, contributing qualitative insights that guided the design and refinement of the Nextgen Classroom dashboard to better address pedagogical and usability challenges. The work carried out in this study focused on the design, development, and implementation of an interactive Learning Analytics dashboard for teachers, as well as its evaluation. Several elements remained out of scope due to time and technical constraints. A student-facing version was not developed, and integration into the company's main application was carried out internally by Nextgen's team because of strict security policies preventing external modifications to their production branch. These limitations were already considered in the scope of our project and left us confined to a teacher-oriented prototype and an independent testing environment.

1.7 Target group

The dashboard project focuses on helping teachers manage diverse student needs and performances. It investigates how effective Learning Analytics dashboards are for

teachers using the Nextgen Classroom web app. By focusing on teachers, the dashboard project aims to provide insights and visualization tools that can directly benefit teachers' way of teaching and consequently improve students' learning. The project's findings and the dashboard itself may also be of interest to educational institutions, instructors, and policymakers who are interested in exploring the potential of Learning Analytics tools to enhance learning outcomes.

1.8 Outline

This study is organized as follows. ***Section 1*** of this study covers the background on educational technology and learning analytics, a review of related work spanning teacher perceptions, language learning challenges, visualization limitations, and co-design approaches, followed by the problem formulation, motivation, scope, and target group. ***Section 2*** establishes the conceptual framework underpinning the study's approach to learning analytics and dashboard design. ***Section 3*** describes the research context and settings, the mixed-methods research design, measures taken to ensure reliability and validity, and ethical considerations. ***Section 4*** details the two core project phases: the creation of a learning analytics dashboard prototype and its subsequent evaluation through participatory workshops. ***Section 5*** presents participant profiles, System Usability Scale (SUS) scores at both overall and item-by-item levels, frequency distributions from post-workshop questionnaires, perceived usefulness ratings, and qualitative findings around key dashboard features, motivators, and barriers. ***Section 6*** interprets the findings by examining usability expectations versus lived experience, an expectation-adjustment effect around perceived usefulness, connections to prior literature, and the study's limitations. ***Section 7*** summarizes the key findings, reflects on their relevance and generalizability, offers a critical reflection on the study, and proposes directions for future study.

2 Theoretical Background

The Technology Acceptance Model is a model initially introduced by Fred D. Davis in 1985 as part of his Ph.D. in Management with the motivation that it would give insight into user acceptance processes and function as a theoretical basis for user acceptance testing. TAM was designed to predict and explain the attitude towards a system as a function of two theoretical constructs, perceived usefulness (PU) defined as "The degree to which an individual believes that using a particular system would enhance his or her job performance" and perceived ease of use (PEU) which is defined as "The degree to which an individual believes that using a particular system would be free of physical and mental effort" [34]. Davis developed and validated six-item scales for the two theoretical constructs in 1989 and found that PU had much higher correlation with predicting the usage of a system [35].

In the context of LA dashboards, TAM is a common framework used to explain users' acceptance toward new dashboards [12], [17], [36], [37]. The most relevant work to our thesis being Chavan et al. who evaluated Tcherly, a teacher-facing LA dashboard using a combination of TAM and System Usability Scale (SUS) [36]. Similarly, this thesis uses a subset of TAM and SUS. More broadly, Scherer et al. conducted a meta-analysis of 114 TAM studies with 34,357 teachers. This concluded that TAM explains the teacher's acceptance of digital learning tools well [37]. This study employs TAM's theoretical construct PU as it directly captures what is measured in the main RQ.

3 Methodological Approach

This section describes the methods used in the design and evaluation of the dashboard and how the results were checked for accuracy. The ethical guidelines followed are also discussed. This research evaluates the effectiveness of a learning analytics dashboard in language teaching through a focus group approach with test users.

3.1 Context and settings

Nextgen Classroom develops educational software designed to support teachers in delivering instruction and to assist students in improving their learning outcomes. Nowadays Nextgen has around 10.000 active users generating data. The platform collects a large amount of it related to teaching activities, student performance, and student interaction within the system. However, this data has not yet been fully utilized to generate insights that could support teachers in monitoring learning progress or adapting their instructional strategies.

To address this limitation, Nextgen Classroom initiated an effort to integrate learning analytics into its platform. The goal of this initiative is to leverage the available data to provide teachers with meaningful insights into student learning processes, particularly in the context of language learning. This thesis was carried out in collaboration with Nextgen Classroom to design and implement a Learning Analytics Dashboard that visualizes students' language-learning progression using data collected by the platform. Through an internship within the company, valuable insights were obtained regarding teachers' pedagogical needs and the functional requirements for the dashboard. The project encompassed a comprehensive analysis of the existing data infrastructure, the iterative design of dashboard mock-ups, and the development of a prototype intended for seamless integration into the Nextgen Classroom environment.

The resulting prototype will be evaluated from the perspective of teachers to assess how effectively it supports the monitoring of student progress and the identification of potential learning difficulties.

3.2 Research methods

This thesis applies a methodological approach combining Design-Based Research (DBR), Co-Design and Participatory Design and a mixed-methods evaluation strategy. These methods were selected because they support the design, development, and evaluation of an educational dashboard while ensuring that it addresses the real needs of its intended users. Together, these approaches allow the research to both develop a practical solution and empirically evaluate its usefulness for teachers in a real educational context.

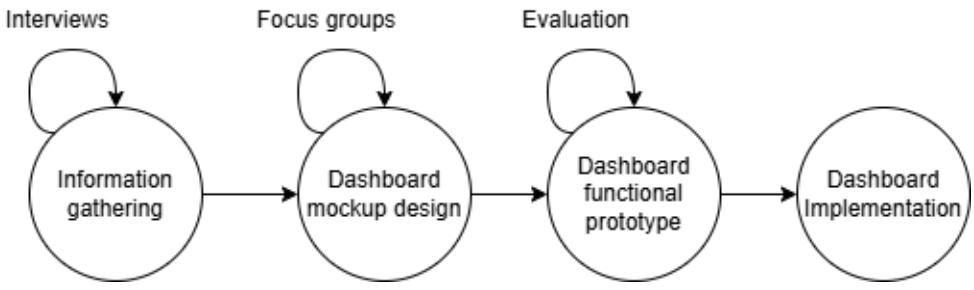


Figure 3.1: Diagram of the process followed [38].

Design-Based Research (DBR)

The primary methodological framework used in this thesis is Design-Based Research (DBR) [38]. DBR is an iterative research approach that combines educational theory with practical development through cycles of design, implementation, evaluation, and refinement. It is particularly suited for EdTech because it ensures that solutions are grounded in real-world teaching and learning contexts [1].

In this project, DBR guides the development of the learning analytics dashboard through four stages: identifying requirements through stakeholder interviews, generating design concepts through brainstorming and low-fidelity prototypes, developing and refining high-fidelity prototypes through collaborative testing, and evaluating the dashboard in an educational context to assess its usability and practical value for teachers [1].

Co-Design and Participatory Design

This project incorporates Co-Design and Participatory Design to involve stakeholders in the development of the dashboard. In this approach, researchers, developers, and teachers act as co-creators throughout the design process rather than merely providing feedback [1], [33], [39]. By involving teachers in activities such as focus groups, interviews, requirement discussions and interface design exploration, the dashboard can better reflect real classroom practices and pedagogical needs [1].

Mixed-Methods Evaluation

The evaluation of the dashboard follows a mixed-methods approach, combining qualitative and quantitative data collection [1], [12], [39]. Qualitative data is gathered through interviews and observations of teachers interacting with the prototype, while quantitative feedback is collected through surveys or questionnaires assessing aspects such as usability, clarity of visualizations, and perceived usefulness in language teaching [12], [39]. Combining these methods enables a more comprehensive evaluation of the dashboard's effectiveness.

Methods Considered but Not Used

Other methodologies were considered during the planning of the study. For example, a long-term field study involving the deployment of the dashboard in real classrooms was considered. Such an approach would provide valuable insights into long-term usage and learning outcomes, but it was not possible to conduct within the scope and timeframe of this thesis.

3.3 Reliability and Validity

To ensure reliability and validity, a set of measures were employed throughout the evaluation process of the proposed Nextgen LA dashboard. First, semi-structured interviews were conducted to gather qualitative feedback on the usefulness and relevance of the dashboard prototype. Second, usability testing through surveys was used to gather quantitative feedback.

3.3.1 Reliability

To ensure reliability, ten participants from different schools participated throughout our interviews and surveys. With this limited sample size, and different groups, the produced results from these groups could differ. To mitigate this, the participants will use the same prototype with the same mock up data to ensure consistent results. Participants answered the same questionnaire and the same interview questions, follow up questions varied between participants which resulted in varying responses.

3.3.2 Validity

The accuracy of the evaluation findings using mock data is limited as the participants were not assessing the dashboard in a real classroom setting with real data. The mock data was used to prevent exposure of real student information.

There is a possibility of bias since four participants out of ten contributed to the design process of this dashboard, and eight out of ten participants have previously used the Nextgen Classroom platform. This could have favourable responses from the surveys and interviews due to the participants' involvement in the development and familiarity with the platform.

Furthermore, the assessment pertains to a prototype implementation, which may exhibit functional limitations or minor inconsistencies that would likely be resolved in the final production version developed by Nextgen Classroom.

Moreover the modification of the scale standard 10-item SUS in the questionnaires introduces a minor threat to validity, as the scale's psychometric properties are formally established for its complete version.

The data collected from participants have not been modified, to ensure integrity of the data and transparency of the results.

3.4 Ethical Considerations

The evaluation required ten participants who were informed of how their data were collected and what it is used for, participation was voluntary, and teachers could withdraw from the evaluation as personal feedback was collected from the participants.

Data were collected directly from participants in the form of an interview or survey and were completely anonymized so participants could not be identified from the results as data protection does not apply to anonymous data. In line with GDPR principles of data minimisation (Article 5(1)(c)) and privacy by design and by default (Article 25), the dashboard was developed and evaluated using mock data to prevent exposure of real student information [40].

4 Research project – Implementation

This study presents the proposed design and prototype implementation of an LA dashboard. Beyond the development phase, a systematic evaluation of the prototype was conducted using established frameworks previously mentioned in section 2.

Accordingly, this chapter is divided into two main sections: the first focuses on the design and implementation of the dashboard prototype, while the second addresses its evaluation and analysis based on empirical data gathered from users.

4.1 Dashboard prototype design and implementation:

Following the process shown in Figure 3.1, the first step was to gather as much information as possible about the Nextgen app and the data used on it. In Figure A.1 of Appendix A, a screenshot of the app itself can be seen. The task was to help Nextgen make effective use of all the data they did not properly show to the teachers, as previously mentioned in section 3.1, the LA dashboard idea was a solution.

After analyzing data from the app, teachers were interviewed and focus groups were conducted to figure out their needs and what are the features they would like to see on the dashboard as well as their personal interests as teachers. For this task requirement engineering [41] (definition in Appendix A) was used to identify some of the following requirements:

- **Compact size view of all statistics**, they wanted to have all the data represented in one place so they don't have to look around through the app to find the info they wanted.
- **View of the full class at the same time**, from there they wanted to select the students they were interested in.
- **Week format statistics**, represent time data in weeks format to see evolution of the students during the semester.
- **Identification of at risk students**, something they needed to know is which students are being left behind and needed more help.

Following the DBR cycle, the design phase involved creating dashboard mockups in Canva [42], which were iteratively brainstormed and refined through analysis, design, implementation, evaluation, and redesign stages. Each iteration integrated the requirements identified earlier and the feedback obtained from teacher focus groups (see planning document in Appendix A), progressively narrowing the design toward what educators found most intuitive and useful. Adjustments included layout organization and selection of graphics, reflecting preferences and usability insights gathered during evaluation. Figure 4.1 illustrates one of the initial iterations where several requirement-driven changes were implemented, while additional versions developed throughout the redesign process are presented in Appendix B.

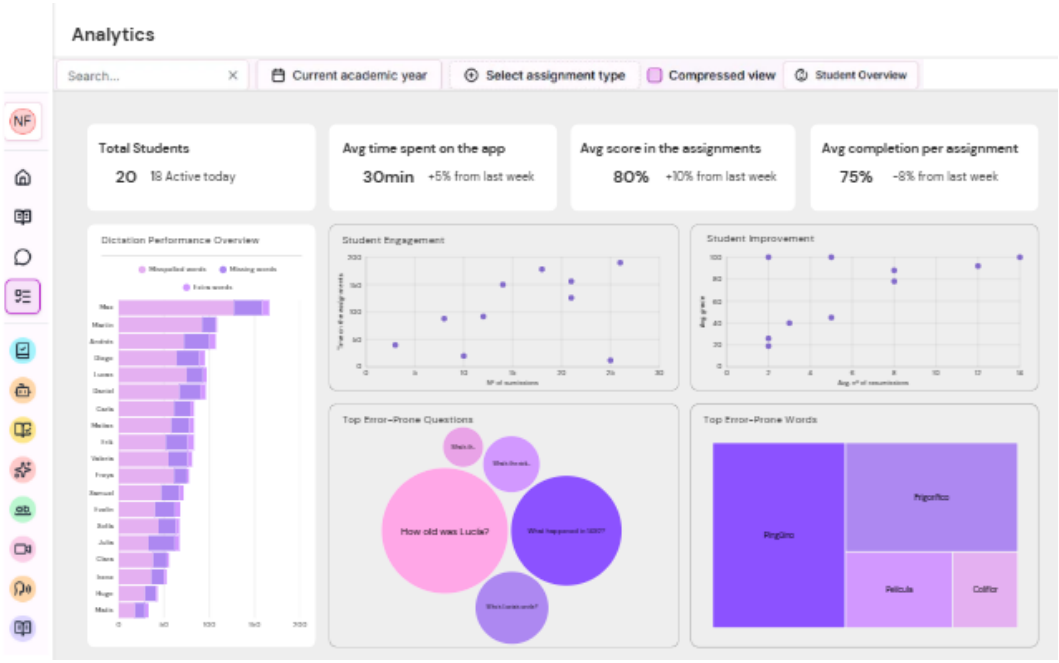


Figure 4.1: First iteration dashboard classview mockup on Canvas.

Figure 4.2 shows one of the later mockups when the final iterations were reached before the dashboard prototype creation. The following changes to the mockup (Figure 4.1) are shown in Figure 4.3. Here can be seen the specific changes done to previously mentioned requirements:

- **Compact size view of all statistics**, the mockups were divided in 2 views one of the individual students and one of the class Figures 4.2 and 4.3.
- **View of the full class at the same time**, class view was created where scatter-plots could represent students as dots in a graph where their performance could be tracked, seen in Figure 4.1.
- **Week format statistics**, individual student graphs show progress over weeks in use time and performance Figure 4.2.
- **Identification of at-risk students**, at the overview level of the class students left behind can be easily identified by the scatter-plot dispersion; they appear at lower levels of engagement and improvement, seen in Figure 4.1.

A prototype was developed in React based on the mockup design (Figure 4.2). The full implementation of the dashboard on Nextgen App it's outside the scope of this project as we explained in section 1.6. Figure 4.3 presents the main interface of the Learning Analytics Dashboard prototype, where aggregated class-level information is displayed simultaneously to support teachers' monitoring and decision-making. The entire prototype is available in the public GitHub repository mentioned in Appendix A, which contains the full source code and a live deployment through GitHub Pages.

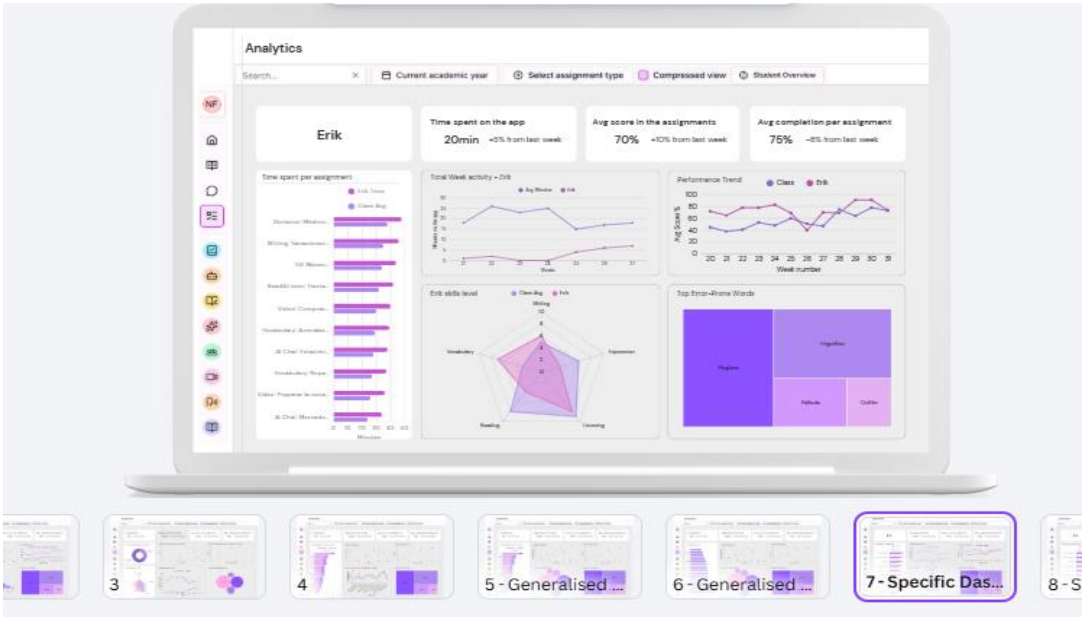


***Figure 4.2**:* Caption of the 7th dashboard specific view mockup on Canvas.

The dashboard was developed as a self-contained (no Nextgen dependencies) prototype using React and TypeScript, Designed to emulate the data structures and interaction patterns of the Nextgen Classroom platform without relying on its internal API. This approach enabled realistic testing in an isolated environment, ensuring that the system remained representative of the production behavior.

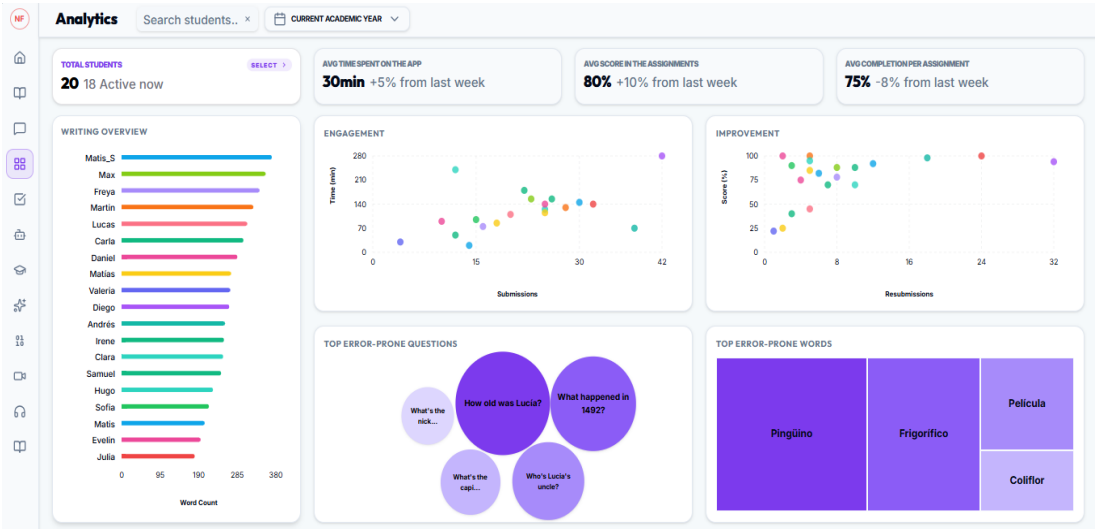


***Figure 4.3**:* Caption of the overview mode of the dashboard prototype.

The unified codebase integrates frontend and backend logic within a modular architecture consisting of three layers: the *Presentation Layer*, the *Logic and Hook Layer*, and the *Data Layer*; as can be seen in Figure 4.4 and further explanation on how the system works and its implementation is provided in Appendix C. The *Presentation Layer (UI)* manages the user interface, layout shells, and interactive controls by rendering modular React components. The *Logic and Pipeline Layer* acts as the application controller, isolating reactive state variables via custom React hooks and utilizing pure

utility pipelines to calculate, transform, and format data for charts. Finally, the *Data Layer* serves as the system's local database, housing statically defined relational JSON collections that act as the single source of data for all operations.

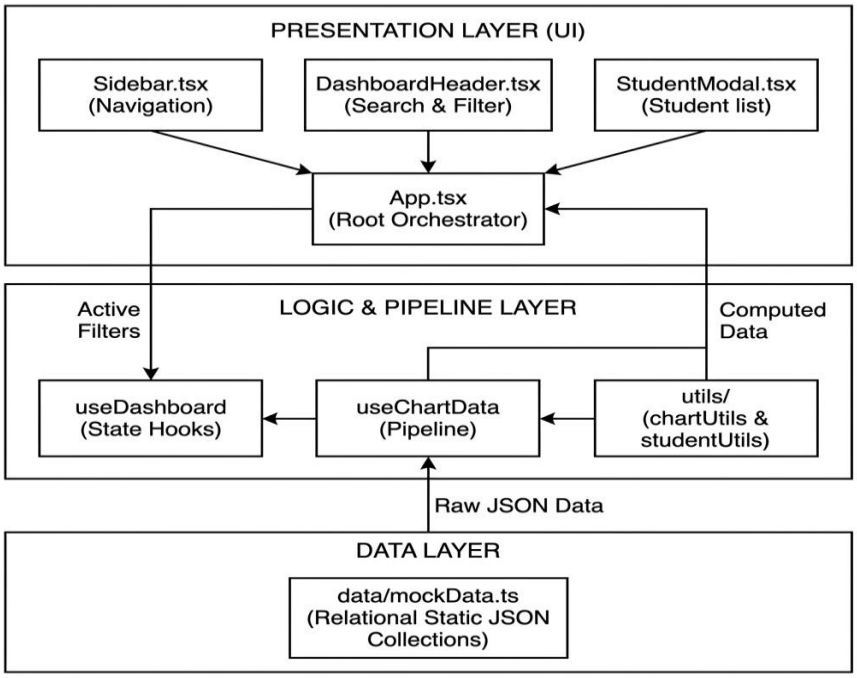


***Figure 4.4**:* Software Architecture of the dashboard prototype.

State management in the Nextgen Dashboard is implemented exclusively using native React Hooks (useState and useMemo) to enhance simplicity and maintainability. Data visualizations rely on two mechanisms: the *Recharts library* for declarative SVG-based charts (bar, line, scatter, radar) and a *custom DOM-based coordinate-packing routine* used to generate the bubble chart by computing explicit positions and sizes applied through inline CSS.

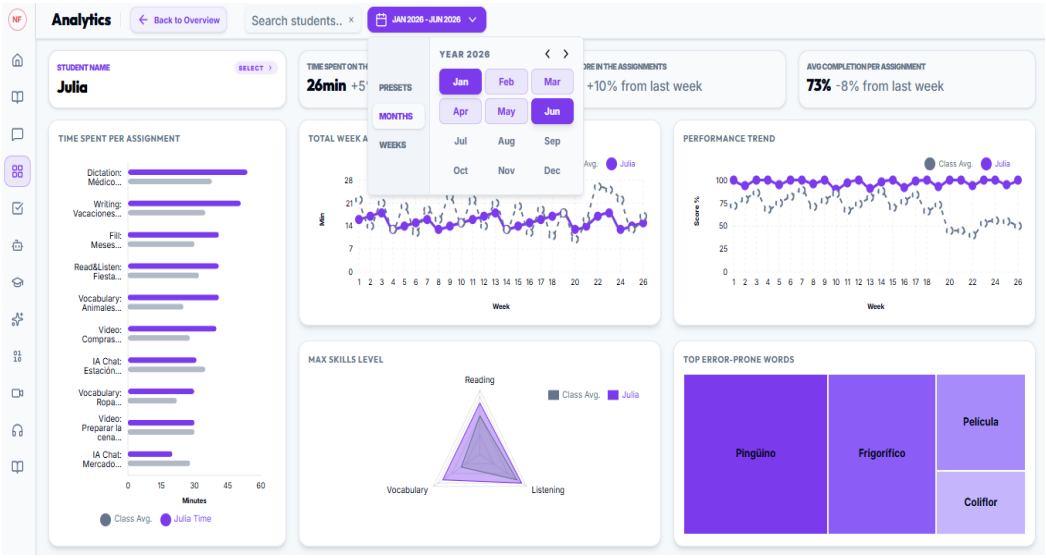


***Figure 4.5**:* Caption of the individual student view for the student Julia.

Figure 4.5 presents the individual student view, using Julia as an example. The time-filtering feature is shown as a calendar at the center of the dashboard. The dashboard shows an overview of the class (Figure 4.3). A teacher could query data on an individual student using the search bar tool (Figure 4.3) showing the student's metrics (Figure 4.5).

4.2 Dashboard prototype evaluation:

After developing the dashboard prototype, the evaluation of its usefulness proceeded in the context of language learning and its overall usability. This evaluation aimed to address the RQs formulated earlier and to analyze the data collected in order to identify potential areas for improvement prior to a full-scale implementation.

To obtain direct and contextually grounded feedback, focus group sessions were held with teachers from various educational institutions. The participants were recruited through contacts provided by Nextgen. This approach allowed for interactive discussions and hands-on testing of the dashboard. The focus groups were conducted at Region Krononberg, Katedralskolan, Linnaeus University, and Mobergsskolan, all located in Växjö and representing different educational levels. Each session followed a structured format consisting of three stages: a pre-questionnaire, a dashboard testing phase, and a post-questionnaire. The pre-questionnaire gathered participant background information and initial expectations; the testing phase involved performing specific tasks (Found in Appendix F) designed to explore all dashboard functionalities; and the post-questionnaire captured participants' reflections and feedback regarding the prototype's usability and pedagogical relevance.

For the design of the questionnaires, a hybrid methodological approach was adopted, integrating items from established frameworks such as the System Usability Scale (SUS) [43], the Technology Acceptance Model (TAM) [35], adapted questions from the Teacher Data Use Survey [44], and questions developed based on data-literacy constructs from the literature of [45]. This combination enabled a comprehensive assessment aligned with our research objectives. The pre- and post-workshop questionnaires results are provided in Appendix D and Appendix E.

As illustrated in Figure A.2 in Appendix A, one of the focus group sessions shows the research team presenting and explaining the dashboard prototype to participants. Each session lasted approximately one hour, and in total, ten participants contributed valuable insights that informed the subsequent refinement of the dashboard design.

5 Results & Analysis

This chapter presents the quantitative and qualitative findings from the pre- and post-workshop questionnaires completed by ten participants. The questionnaires were structured around two theoretical constructs: usability (measured with a modified version of the System Usability Scale (SUS), with adjusted scores to maintain comparability) and perceived usefulness (measured with a five-item Likert scale). Open-ended questions on desired features and motivators/barriers complement the Likert data, providing feedback on what teachers expect from a language learning dashboard and what would encourage them to use it regularly, which quantitative questions could not capture.

All Likert responses use a 1–5 scale (1 = strongly disagree / never, 5 = strongly agree / always). The SUS produces a composite score between 0 and 100 using the standard algorithm; scores in the range 41–60 are conventionally classified as 'OK', 61–80 as 'Good', and above 80 as 'Excellent'. Given the small sample (n = 10), no parametric tests are applied; a Wilcoxon signed-rank test was used for paired pre/post comparisons, though results must be interpreted with caution.

5.1 Participant Profile

All ten participants completed both questionnaires, providing a complete matched dataset. The group was experienced: six held more than 20 years of teaching experience, three had 11–20 years, and one had 6–10 years. The majority (six) rated their digital tool skills as Advanced, with four rating themselves Intermediate. Roles were diverse: five IT-pedagogues, three secondary school teachers, and two university teachers. Eight of ten had previously used the Nextgen platform, whereas eight had never actively used a learning analytics dashboard (though six were aware of them). Related graphs can be seen in the Appendix D. This profile indicates a digitally confident group with limited prior exposure to learning analytics tools specifically.

5.2 System Usability (SUS)

5.2.1 Overall SUS Scores

SUS scores were calculated for each participant using the standard formula [43]. The formula involves summing the adjusted responses for each of the ten items (where positive statements are scored as the response minus one and negative statements as five minus the response) and then multiplying the total by 2.5 to obtain the final usability score. Since only nine SUS questions were used in our questionnaire, the final score was proportionally adjusted using a 10/9 scaling factor to maintain comparability with the standard ten-item SUS. The question addressing the system's inconsistency was removed because it was considered too complex for teachers to interpret ("I thought there was too much inconsistency in this system."). Additionally, the total number of questions was limited to maintain engagement and prevent respondents from losing interest in the questionnaire. Furthermore, another item ("I found the system very cumbersome to use.") was rephrased into a simpler and more accessible version for teachers: "Very unmanageable."

Measure	Pre-workshop (Expectation)	Post-workshop (Experience)
Mean SUS score	47.6 / 100	50.3 / 100
SUS classification	OK	OK

***Table 5.1**: SUS score calculations from the questionnaires.*

Both the expected and experienced SUS scores fall in the 'OK' category (41–60). The post-workshop mean (50.3) is marginally higher than the pre-workshop expectation (47.6), suggesting that actual hands-on experience with the dashboard was slightly more positive than anticipated. However, the Wilcoxon signed-rank test yields a p-value of 0.359, indicating that this difference is not statistically significant at any conventional threshold. Given the small sample, this result must be treated as inconclusive rather than evidence of no effect.

5.2.2 SUS Item-by-Item Analysis

Table 5.2 shows the mean rating for each of the nine SUS items, comparing pre-workshop expectations with post-workshop experience. Items marked [neg] are negatively worded; lower scores on these items indicate better usability.

SUS Item	Pre Mean	Post Mean	Change
Would use frequently (Q7/Q8)	2.50	3.30	0.80
Unnecessarily complex (Q8/Q9) [neg]	3.10	2.80	-0.30
Easy to use (Q9/Q10)	3.50	2.90	-0.60
Need technical support (Q10/Q11) [neg]	2.30	2.00	-0.30
Functions well integrated (Q11/Q12)	4.00	3.20	-0.80
Most people learn quickly (Q12/Q13)	3.50	3.30	-0.20
Very unmanageable (Q13/Q14) [neg]	2.10	1.90	-0.20
Felt confident using (Q14/Q15)	3.80	2.60	-1.20
Needed to learn a lot (Q15/Q16) [neg]	2.70	2.50	-0.20

***Table 5.2**: SUS mean category and change from the questionnaires.*

Several patterns are noteworthy. The item 'Would use frequently' increased from 2.50 to 3.30, suggesting that hands-on exposure raised participants' intention to use the system. Conversely, 'Easy to use' decreased slightly (3.50 to 2.90), and 'Felt confident using' dropped from 3.80 to 2.60. This indicates that while participants had optimistic expectations about ease of use and confidence, the actual experience revealed more friction than anticipated. The negative items 'Unnecessarily complex', 'Need technical support', and 'Very unmanageable' all decreased after the workshop, which is a positive signal, participants found the system less complex and unmanageable than feared.

5.2.3 Frequency Distributions (Post-Workshop Items)

The following tables show the distribution of responses for three key post-workshop SUS items.

On intention to use frequently (Figure E.1), four out of ten participants gave a rating of 4, and one gave 5, while three rated it 2. Perceived ease of use (Figure E.3) was concentrated at 3 (four participants), with no participant rating it 5, suggesting moderate rather than strong ease-of-use. Confidence (Figure E.8) was spread across all five points, with no majority position, reflecting meaningful individual differences in how comfortable participants felt.

5.3 Perceived Usefulness

Six items assessed perceived usefulness, that is, participants' belief that the dashboard would support their professional practice.

Item	Pre Mean	Post Mean	Change
Improve job performance	3.70	3.40	-0.30
Useful in job	3.80	3.20	-0.60
Correctly interpret visualisations	3.80	3.50	-0.30
Translate insights to classroom actions	4.00	3.30	-0.70
Identify students needing support	4.20	3.70	-0.50
Confident drawing conclusions from data	3.80	3.60	-0.20

***Table 5.3**: Usefulness items mean category and change from the questionnaires.*

A counterintuitive finding emerges: all six perceived usefulness items show a decrease from pre- to post-workshop. The largest drops are on 'Translate insights to classroom actions' (4.00 to 3.30) and 'Useful in job' (3.80 to 3.20). The Wilcoxon test for the mean

PU score yields p = 0.26, so the decrease is not statistically significant; however, the consistent downward direction across all six items is noteworthy. Analyzing individual decimal drops (e.g., from 4.00 to 3.30) in a small sample (n = 10) is highly speculative and should be viewed as qualitative observations or tentative hypotheses for future research, rather than concrete trends.

Moreover, looking into the change in standard deviation, there are only two items whose difference exceeds 0.2, which might be worth noting because they represent a lack of consensus after the workshop (*Standard deviations for individual items ranged from approximately 1.0 to 1.5 across both conditions)*. This could suggest that teachers' opinions about the usefulness of the dashboard differed following the session, indicating that participants had more varied perceptions of these aspects after engaging with the dashboard.

Item	Pre SD	Post SD	Change
Improve job performance	1.06	1.26	0.21
Useful in job	0.92	1.40	0.48

Table 5.4*: Usefulness items standard deviation and change from the questionnaires.*

This pattern may reflect an 'expectation adjustment' effect: before the workshop, participants held relatively optimistic expectations about what a learning analytics dashboard could offer, but after interacting with the actual system, they moderated those expectations. Crucially, all post-workshop means remain above the midpoint of 3.0 on a 5-point scale, indicating that the dashboard was still perceived as useful just less so than initially hoped.

The highest-rated post-workshop item is 'Identify students needing support' (3.70), suggesting that this specific functionality resonated most clearly with participants' professional needs.

5.3.1 Frequency Distributions (Post-Workshop Usefulness Items)

On job performance improvement (Figure E.10), no participant selected 1 or 3, opinions were polarised between moderate (2, four participants) and strong (4 and 5, six participants) agreement. On usefulness in job (Figure E.11), responses are broadly spread, reflecting genuine diversity in how relevant participants found the dashboard to their specific role. Interpretation of visualisations (Figure E.12) was the most positively rated post-workshop item, with six of ten participants selecting 4, which aligns with participants' generally advanced digital skills.

5.4 Qualitative Findings

5.4.1 Most Important Dashboard Features (Q22/Q23)

Participants were asked to list their three most important expected features both before and after the workshop. Across both questionnaires, five themes dominated:

- **Ease of use and navigation**: mentioned in nearly every response, both pre and post. Typical phrasings included '*easy to use*', '*easy onboarding*', and '*easy to navigate'*.
- **Student progress tracking:** participants wanted to follow individual and group development over time, both in terms of performance and effort.
- **Clear visualisation and data interpretation**: responses emphasised that data should be easy to read and conclusions easy to draw without extensive manual analysis.
- **Individual-level data**: the ability to view and act on data at the individual student level was ranked high, often linked to identifying students needing support.
- **Relevance and criteria alignment**: particularly among university teachers, there was a desire for data that relates to learning goals and assignment criteria rather than only time-on-task or quantitative metrics.

After the workshop, one recurring addition was system integration: two participants noted that they would only adopt the dashboard if it were connected to the tools they already use (Classroom, Google, Glosor.eu), noting that having to re-enter everything in Nextgen would be a barrier.

5.4.2 Motivators and Barriers (Q23/Q24)

Open-ended responses on motivators and barriers were consistent across both questionnaires and cluster into the following themes:

Motivators: Ease of use was the single most mentioned motivator. Decision support (the ability to quickly see which students need attention) was cited as a compelling driver. Student agency was mentioned by one university teacher, who noted that the system would be more motivating if students could also access and use their own data. A good overview and fast access to relevant information were also highlighted.

Barriers: Complexity and difficult navigation were the most frequently cited barriers. Requiring many clicks to reach information was singled out by one participant. Data quality and reliability were raised as concerns, if teachers cannot trust the data, they will not use the dashboard. Data integration challenges (needing to import or manually enter data from other systems) were mentioned as a significant practical obstacle. Privacy was identified as a potential barrier, but it has been a core design constraint throughout the project.

6 Discussion

This chapter discusses the findings in relation to the research questions, compares them with prior work in the field of learning analytics, and considers the impact of limitations on how the results can be generalised.

6.1 Usability: Expectations vs. Experience

Pre- and post-workshop mean SUS scores (47.6 vs. 50.3, both classified as '*OK*') suggest that the dashboard broadly met participants' usability expectations without dramatically exceeding or disappointing them. This is a nuanced finding. On one hand, a score in the '*OK*' band falls short of the 'Good' threshold (61–80) typically associated with broad adoption in professional settings. On the other hand, participants arrived with prior experience of Nextgen (80% had used the platform) and moderate-to-advanced digital skills, meaning their expectations were already reasonably calibrated.

Participants expected higher ease of use and confidence (Q9, Q14) than they experienced. This gap between expectation and experience is a known phenomenon in technology adoption research. In the context of learning analytics, [46] found that teachers often perceive such tools as useful in principle but encounter usability friction when interacting with specific implementations. The current findings align with this pattern.

At the same time, the post-workshop reduction in the negative items (complexity, unmanageability, need for technical support) indicates that the dashboard performed better than feared in those respects. Participants had expected a steeper learning curve than they encountered (Q15), which is a positive signal for adoption. To address RQ1a, after participants hands-on interaction with the dashboard, teachers perceived usability as ‘OK’ with a SUS score of (50.3). Ease of use expectations scored higher than the experience which resulted in a mean score of (2.90), putting it just below midpoint of (3.0) on a 5-point likert scale.

6.2 Perceived Usefulness: The Expectation-Adjustment Effect

The consistent decrease across all six perceived usefulness items is a notable pattern of this study. This pattern is consistent with the technology acceptance literature, where pre-use optimism is a common feature of teacher attitudes toward EdTech [47]. After hands-on interaction, users develop a more grounded and sometimes more critical assessment. Analyzing this decrease in a small sample (n = 10) is highly speculative, and should be viewed as qualitative observations or tentative hypotheses for future research, rather than concrete trends.

Importantly, all post-workshop means remain above 3.0, meaning participants still perceived the system as useful, the adjustment was one of moderation, not rejection. The highest post-workshop usefulness item, '*Identify students needing support'* (3.70), points to a concrete professional need that teachers believe the dashboard can address. This aligns with research by [48], who found that teachers are most receptive to learning analytics tools when they perceive a direct link between the data and actionable decisions about individual students.

The largest decreases occurred on 'Translate insights to classroom actions' (−0.70) and '*Useful in job*' (−0.60). The qualitative data help explain this: participants noted that the current dashboard primarily provides quantitative metrics (time on task, activity frequency) but does not show how students perform relative to learning objectives or assignment criteria. One teacher noted that the system showed activity well, but could not tell how the students performed against the course or specific assignment. This distinction between usage data and learning quality data is a fundamental tension in learning analytics design [25], and it appears to have influenced post-workshop assessments directly.

To address RQ1b, after participants hands-on interaction with the dashboard, interpretation of the visualizations scored a mean (3.50), and the perceived usefulness of its visual representations scored means (3.2 - 3.7) putting both above midpoint of (3.0) on a 5-point likert scale. Every item scored lower than the expectation, although the Wilcoxon test yielded $p = 0.26$, showing that the decrease is not statistically significant.

6.3 Qualitative Themes and Their Relation to Prior Work

The qualitative findings reinforce results from the broader literature. The common emphasis on ease of use as a requirement for teachers to adopt the dashboard matches findings across numerous EdTech studies [35], [37]. The demand for individual-level student data aligns with research showing that teachers' primary professional concern is the progress of specific students rather than aggregate class trends [49].

The concern about data relevance, specifically that the dashboard should show student performance against learning criteria, not only activity data reflects a broader critique in the learning analytics literature [25]. argue that most current learning analytics systems capture what is easy to measure rather than what is educationally meaningful. The participants in this study appear to share that concern.

System integration emerged as a post-workshop barrier not strongly anticipated beforehand. Participants who rely on Google Classroom, Glosor.eu, and other platforms expressed reluctance to duplicate their workflow in Nextgen. This echoes the [50] 'learning analytics cycle' argument that data collection must be embedded in existing teaching practice to be sustainable. A dashboard that requires parallel data entry could increase friction in teachers' adoption regardless of its analytical capabilities and would benefit from connecting to platforms already used by teachers.

The privacy concern raised by one participant [40] is consistent with the literature on teacher attitudes toward data-driven tools in European educational contexts [51]. Though mentioned by only one participant, it may represent a concern shared more broadly.

To address RQ1c, the qualitative findings revealed ease of use, easy navigation, visualizations that are easy to interpret without requiring a manual to be the most dominant design factors. Student progress tracking, learning goals, assignment criteria dominated as the data factors, and integration with common teaching tools such as Google Classroom and Glosor.eu dominated as the integration factor.

6.4 Limitations and Their Impact

Several limitations constrain the conclusions that can be drawn from this study, impacting the generalizability of the findings and informing the direction of future research.

First, the study relies on a small (n = 10) and highly homogeneous sample. The participants consisted of five Swedish language educators and five IT-pedagogues attending the same workshop, and all but one were 45 or older. Furthermore, there is a high risk of favorable participant bias, as four of these individuals actively contributed to the dashboard's design process, and eight had prior experience using the Nextgen platform. This lack of diversity severely limits the generalizability of the findings. To establish a more representative picture of teachers perceived usefulness and usability of the dashboard, it directly necessitates future studies involving broader participant samples, specifically targeting teachers with lower digital confidence and at different career stages.

Second, the quantitative data collected via Likert scales did not yield statistically significant differences, which limits the strength of the conclusions. Grounding this in the data, the Wilcoxon signed-rank tests resulted in a p-value of 0.359 for the System Usability Scale (SUS) scores and a p-value of 0.26 for the perceived usefulness scores. While this could reflect a true null effect, it may also indicate insufficient statistical power due to the small sample size.

Third, the questionnaire design, while grounded in established instruments (SUS, TAM-inspired usefulness items), did not capture certain dimensions that emerged as important in the qualitative data, particularly data quality, criteria alignment, and workflow integration, so these dimensions remain unquantified. Including validated subscales for these constructs in future studies would strengthen the measurement framework.

Finally, the evaluation is constrained by methodological and technical limitations regarding the prototype itself. Participants evaluated the system using mock data rather than live student data in a real classroom setting, limiting the ecological validity of the test. The mock data was representative of a class and the usability and usefulness would likely be similar. Additionally, due to company security considerations and time constraints, the dashboard was not integrated into the main Nextgen Classroom platform; instead, it was evaluated as an independent, self-contained prototype built using React and TypeScript. This prototype may have exhibited functional limitations or minor inconsistencies not present in a final production version, which could have negatively influenced users' initial impressions. This technical constraint highlights the importance of system integration as a key priority for future work, ensuring the dashboard can seamlessly connect with existing platforms without requiring parallel data entry.

7 Conclusions and Future Work

All these conclusions should be taken carefully with the limitations commented in section 6.4 in mind.

7.1 Summary of Findings

The main findings are as follows:

- **Usability** was rated in the '*OK*' band both before and after the workshop, with no statistically significant change. Participants found the system less complex than anticipated but also less easy to use and confidence-inspiring than they had hoped.
- **Perceived usefulness** decreased across all six items from pre- to post-workshop, though all post-workshop means remained above the scale midpoint. This may suggest an expectation-adjustment effect rather than a rejection of the dashboard's potential.
- **Qualitative data** consistently identified ease of use, clear data visualisation, individual student tracking, and integration with existing tools as the most critical requirements.
- **Key barriers**: complexity of navigation, limited coverage of qualitatively meaningful data (performance against learning criteria), and the burden of entering data into a separate platform.

In relation to the initial research problem, whether teachers can effectively use a learning analytics dashboard to support language instruction. The data suggest a cautiously positive but conditional answer. Teachers expressed genuine interest and identified specific use cases (particularly identifying students who need support). The dashboard scored 'OK' and met usability expectations of the participants (RQ1a). After participants' hands-on experience with the dashboard, their score on interpretation of the visualizations decreased slightly, and so did the perceived usefulness of its visual representations. Despite this decrease, both scores were positive (RQ1b). However, the current implementation does not fully meet their needs for relevant, easy-to-access, criteria-aligned data and integration with existing teaching tools (RQ1c).

7.2 Relevance and Generalisability

The findings are most directly relevant to developers and administrators of learning analytics platforms in K–12 and higher education language contexts, particularly in Swedish or similar northern European educational systems where Nextgen is used. More broadly, the emphasis on ease of use, integration, and data relevance echoes challenges reported across many EdTech deployments, suggesting that the findings have some transferability to similar contexts.

Although based on a small sample, the results are relevant as they provide evidence-based insight into teachers' current readiness to use learning analytics and the practical conditions that shape adoption. The results show that perceived usefulness declined after practical use, highlighting that aligning teachers' expectations with the actual capabilities of a dashboard is an important design and implementation consideration. This effect emerged independently of usability scores, which remained stable.

7.3 Future Work

Several directions for future research and development follow from these findings:

- **Longitudinal evaluation**: Conduct a follow-up study measuring actual dashboard usage and updated perceptions after one or more school terms, to assess whether initial impressions predict sustained adoption.
- **Criteria-aligned analytics**: Investigate whether and how learning analytics dashboards can be extended to capture qualitative performance data linked to language learning objectives, addressing the most frequently cited gap in current functionality.
- **Practical feedback**: Explore how learning analytics dashboards can provide actionable feedback instead of raw performance data, in order to help teachers identify where students struggle.
- **System integration**: Explore technical pathways for connecting Nextgen with other platforms teachers currently use (Google Classroom, dedicated vocabulary tools), so that the dashboard can aggregate data without requiring separate data entry.
- **Broader participant samples**: Replicate the study with secondary school language teachers who have lower digital confidence, and with teachers at different career stages, to establish a more representative picture of dashboard readiness across the profession. This is necessary as the current group is predominantly composed of highly experienced IT-pedagogues who do not reflect the broader teaching profession, which could lead to a different result.
- **Student-facing analytics**: One participant raised the possibility of students accessing and using their own learning data. Future work could examine the design and effectiveness of student-facing dashboards as a complement to teacher-facing tools.
- **Improved measurement instruments**: Future iterations of this study should create or integrate validated subscales that capture dimensions such as data quality, criteria alignment, and workflow integration. These dimensions were not measured by the SUS and TAM questionnaires but arose from the qualitative findings.

In conclusion, this study shows that teachers can use the dashboard and see potential value in it, but their overall experience depends strongly on the relevance of the data provided and how well the tool fits into their existing workflows. Usability was acceptable, yet the decline in perceived usefulness after hands-on use. The current version broadly meets teachers' instructional expectations, though there remains room for refinement. The qualitative findings reinforce this by highlighting the importance of clearer visualisations, criteria-aligned information, and better integration with existing systems. These results do not offer definitive claims about the broader effectiveness of learning analytics dashboards, but they do identify concrete areas where design and implementation need improvement to better support language teachers in practice.

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A Appendix: Project Images

Here the images and tables from the project in general can be found. The source code can be found on my github repository: [Axton-Industries/Nextgen-work: Here you can find the work I've done for the Nextgen App.]

Moreover here can be found the diary where feedback and work done during the Nextgen internship has been reflected: [Planning document Adrian]

Requirements engineering (RE) is a critical phase in software development that ensures the final product meets the needs of its users and stakeholders. It involves a series of activities aimed at understanding and defining what stakeholders require from the system, ensuring that these requirements are met throughout the project lifecycle.

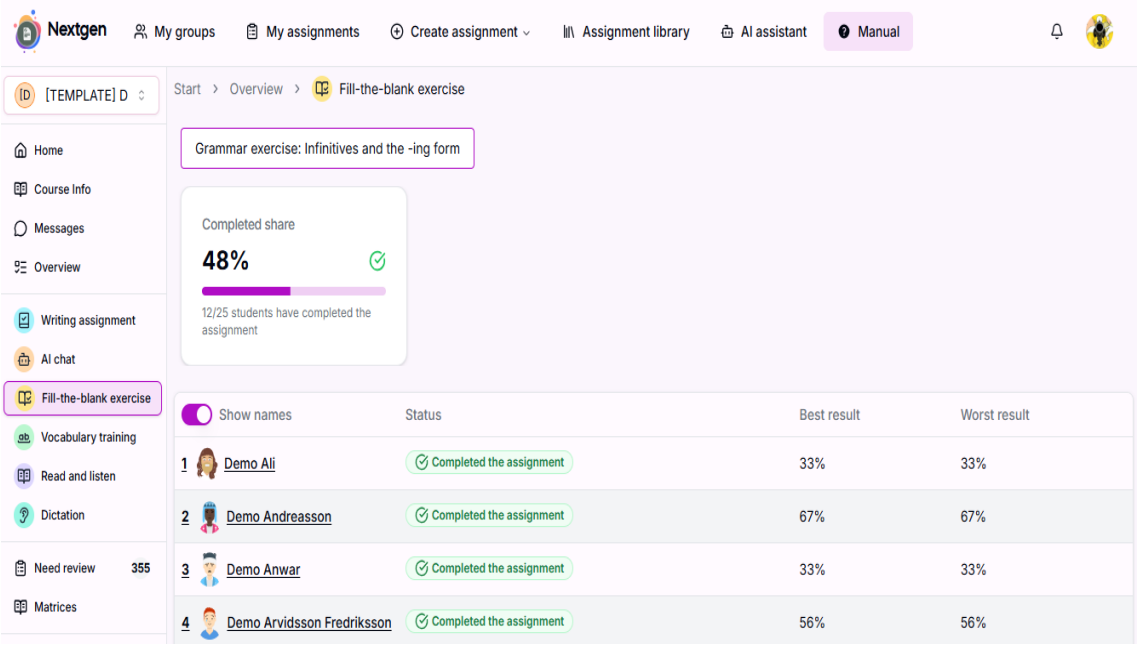


Figure A.1*:* Caption of a Demogroup inside Nextgen App with analytics data.



Figure A.2*:* Picture of the focus group at Region Kronoberg.

B Appendix: Project mockups

Here can be found the mockups iterations made on Canva previous to the prototype creation.

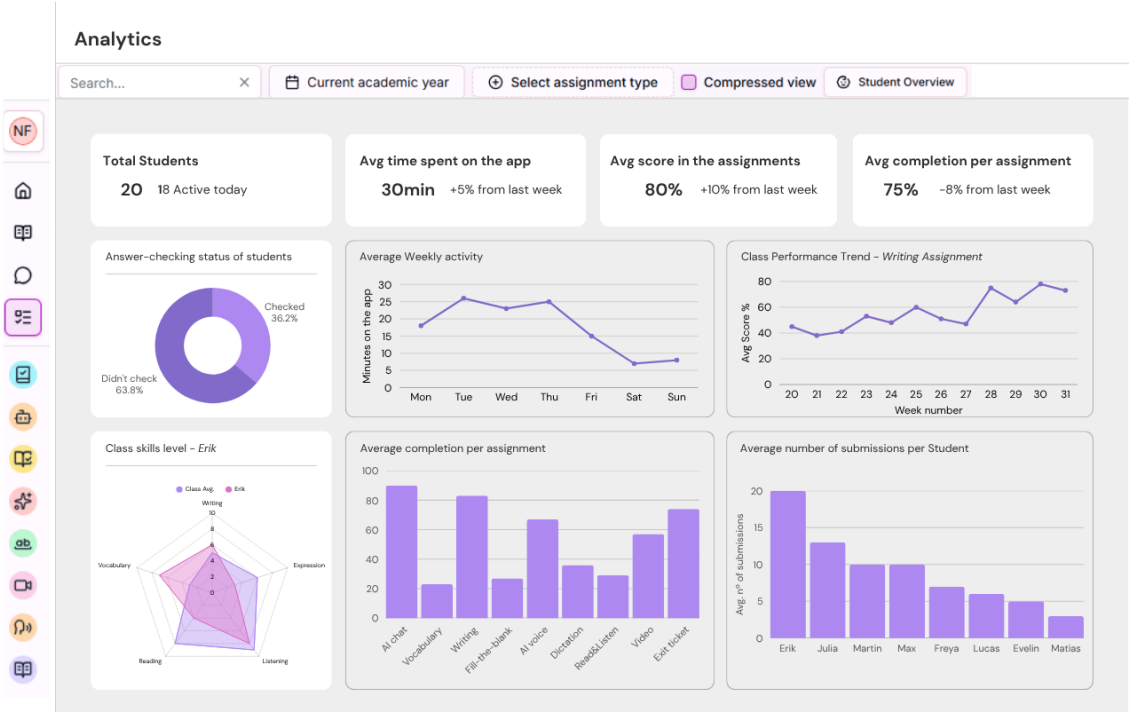


Figure B.1: App mockup version 1.

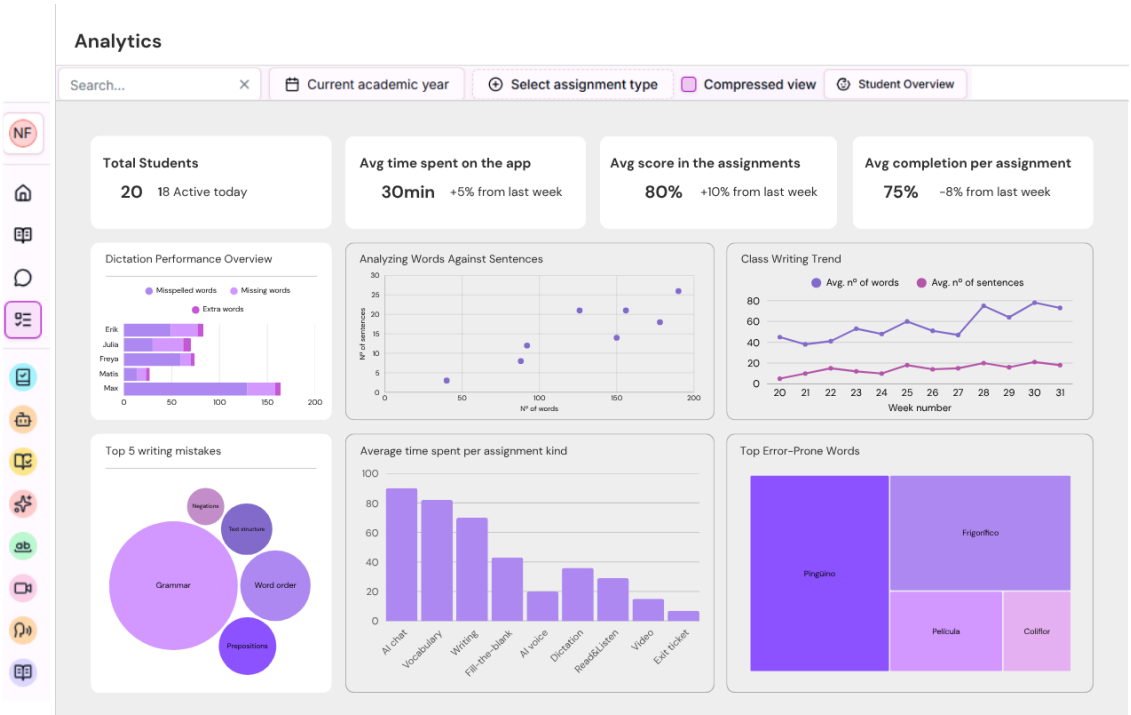


Figure B.2: App mockup version 2.

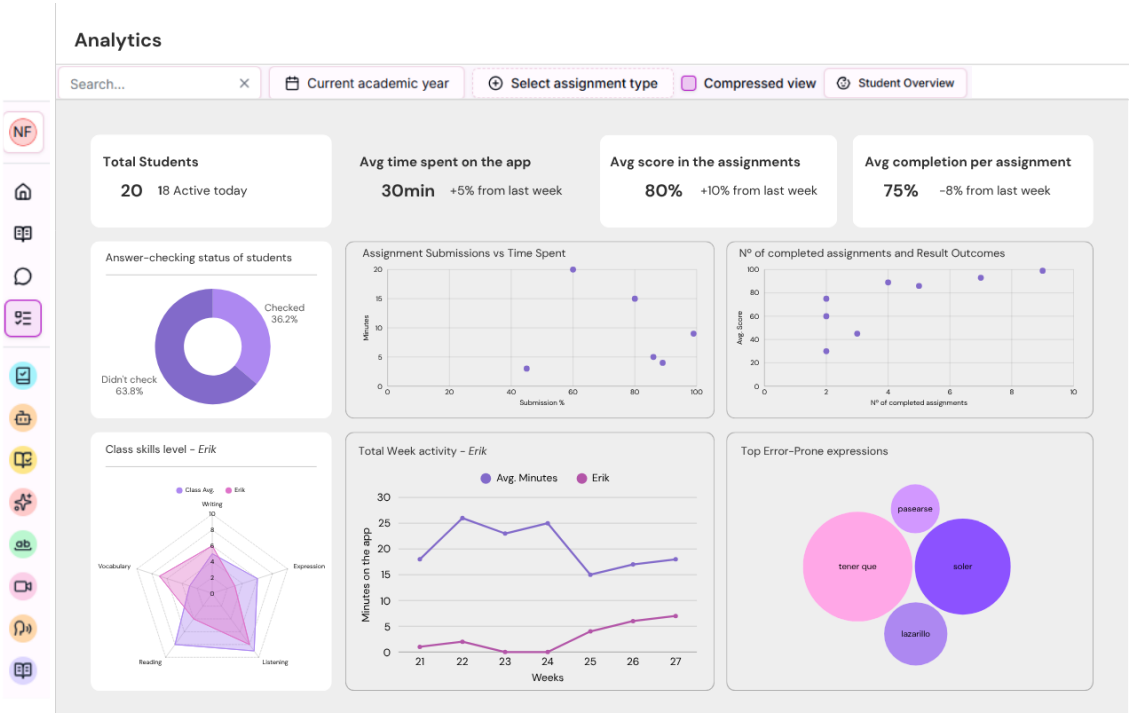


Figure B.3: App mockup version 3.

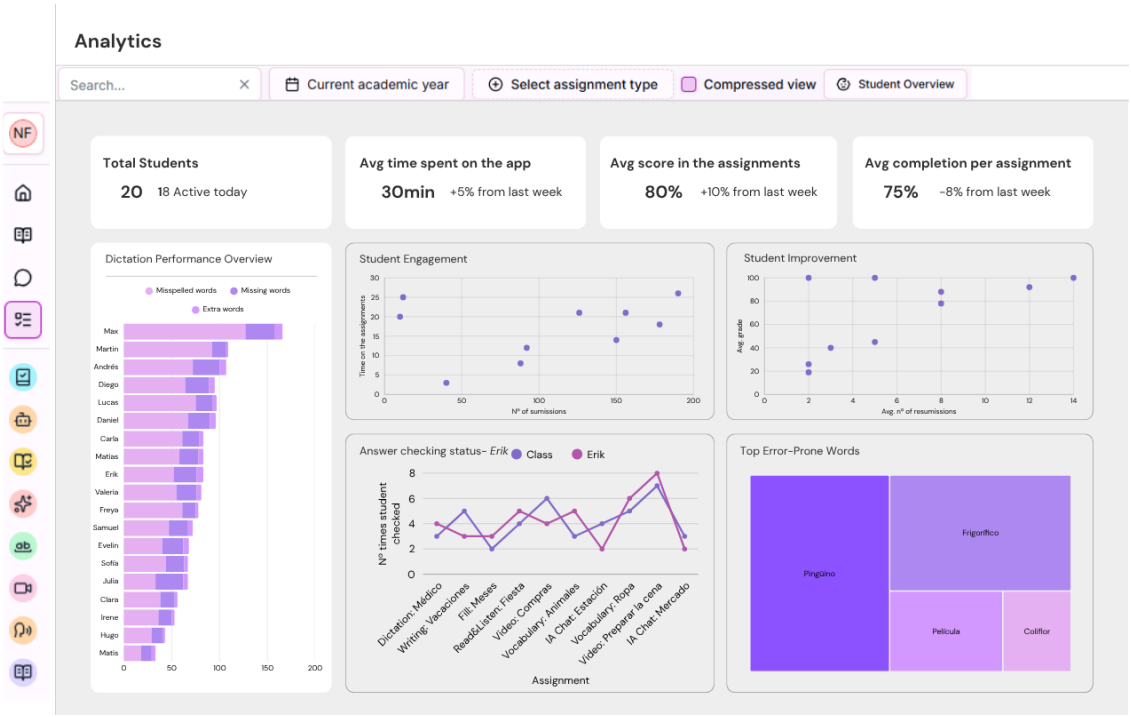


Figure B.4: App mockup version 4.

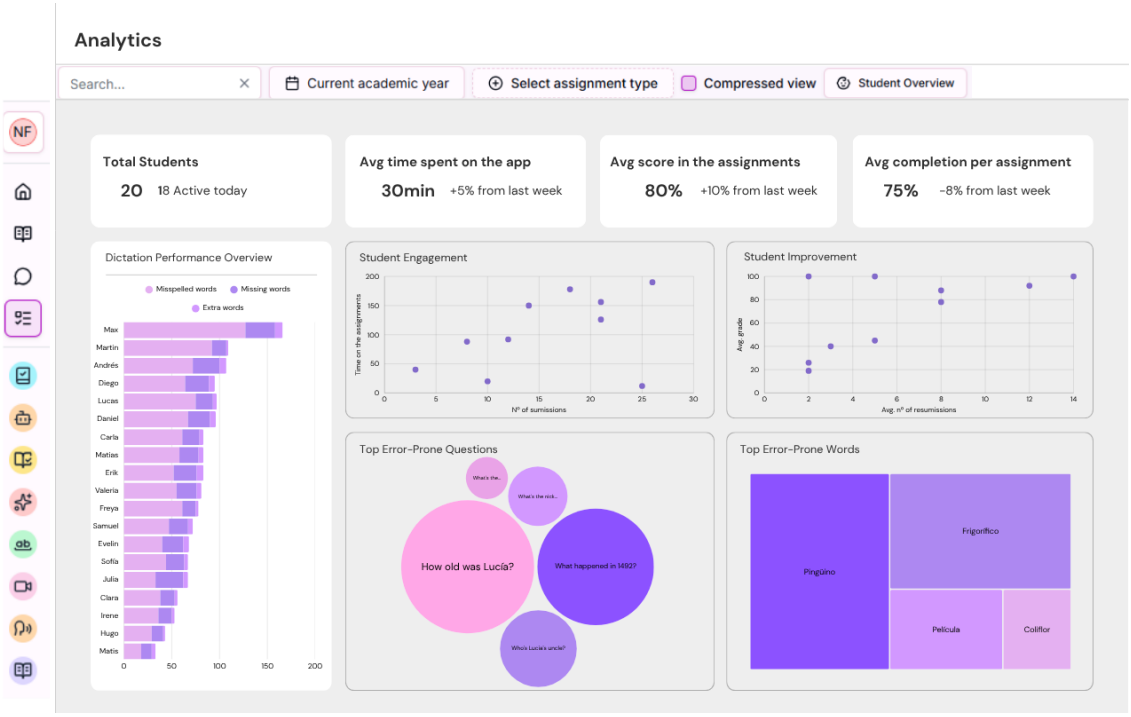


***Figure B.5**: App mockup version 5.*

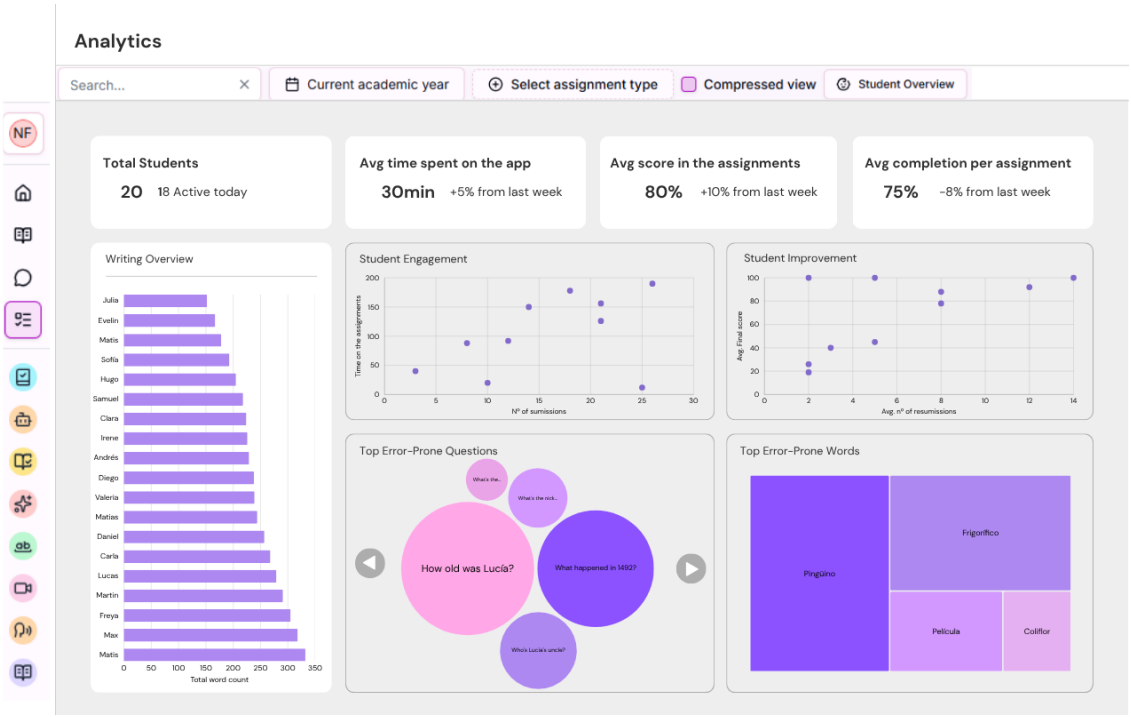


***Figure B.6**: App mockup version 6.*

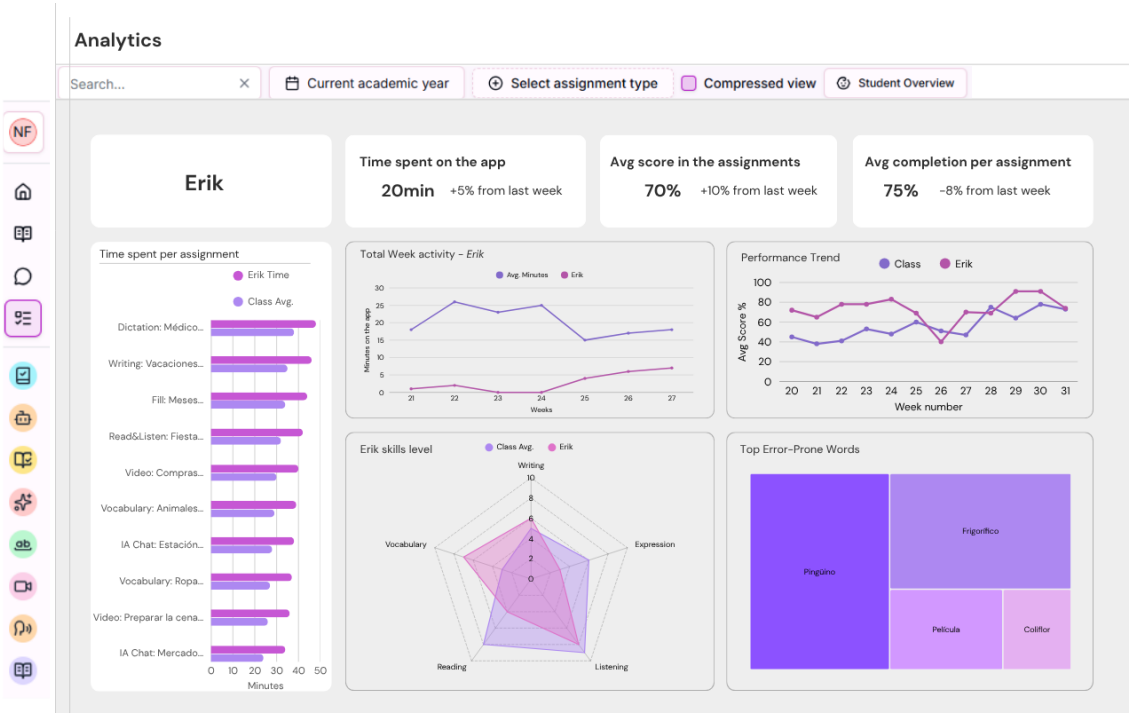


***Figure B.7**: App mockup, individual view version 1.*

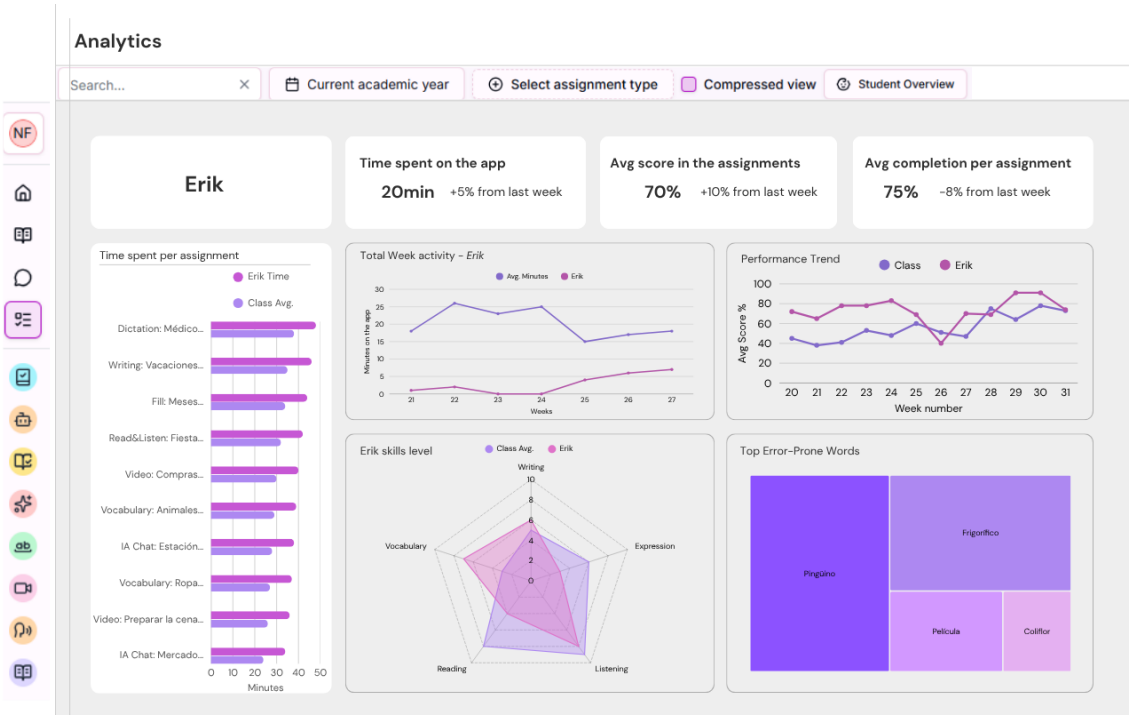


***Figure B.8**: App mockup, individual view version 2.*

C Appendix: Implemented Architecture

Here the UML graphs can be found that explain deeper how our software architecture works for our designed dashboard prototype in React and a more profound understanding of the libraries used.

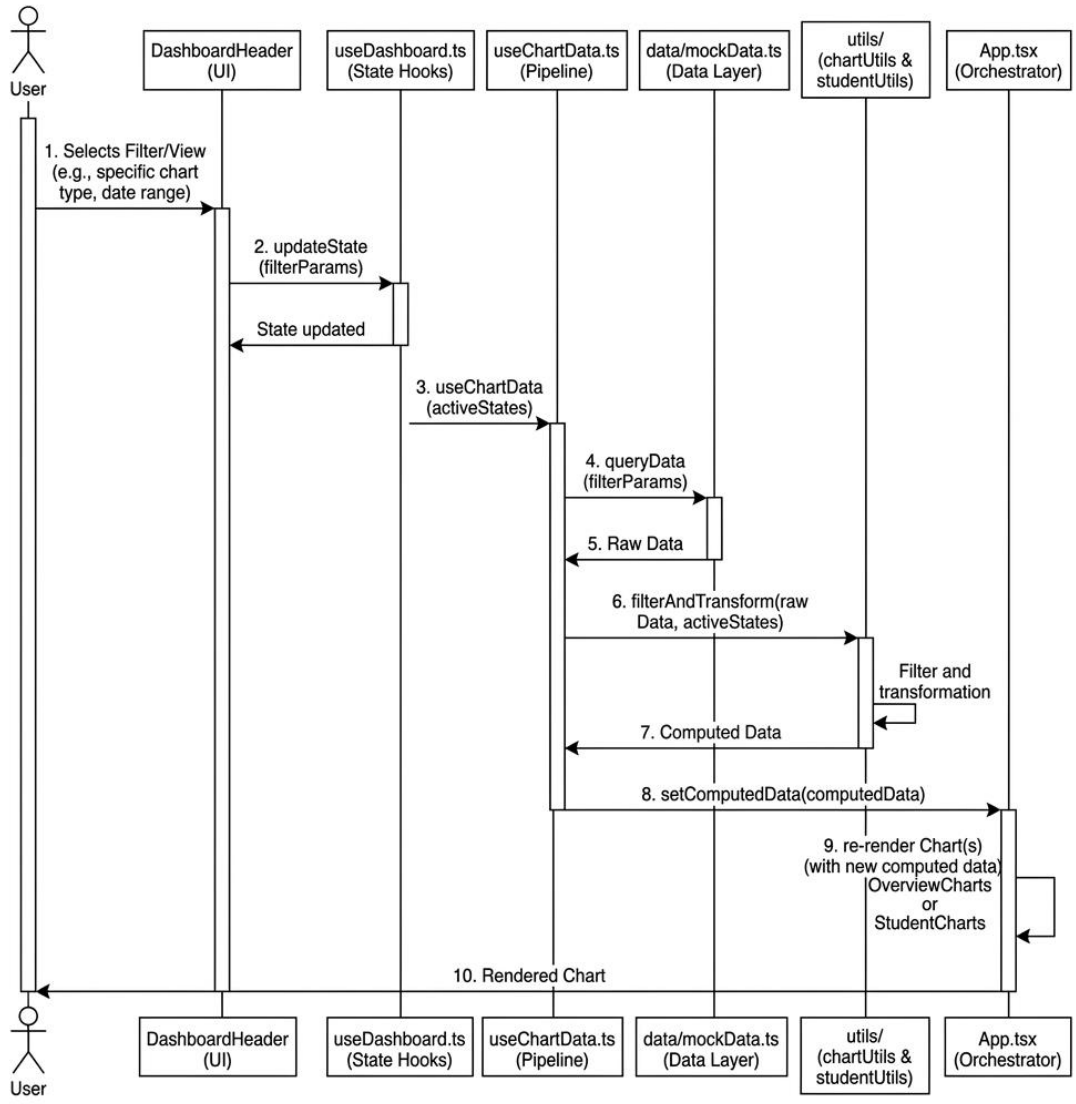


***Figure C.1**:* UML Sequence Workflow Diagram of the dashboard.

Figure C.1 illustrates the sequential workflow of the dashboard's interactive logic. When a user performs an action (such as applying a filter or selecting a student profile) the event triggers state updates within useDashboard.ts. These updated parameters propagate to useChartData.ts, which processes the raw records stored in data/mockData.ts and applies transformations through helper utilities (chartUtils.ts and studentUtils.ts) to compute colors and bubble coordinates. The resulting metrics are then passed back to App.tsx, prompting a reactive re-render of the relevant charts. This flow demonstrates

how user input cascades through state management, data transformation, and visualization layers to produce dynamically updated charts in real time.

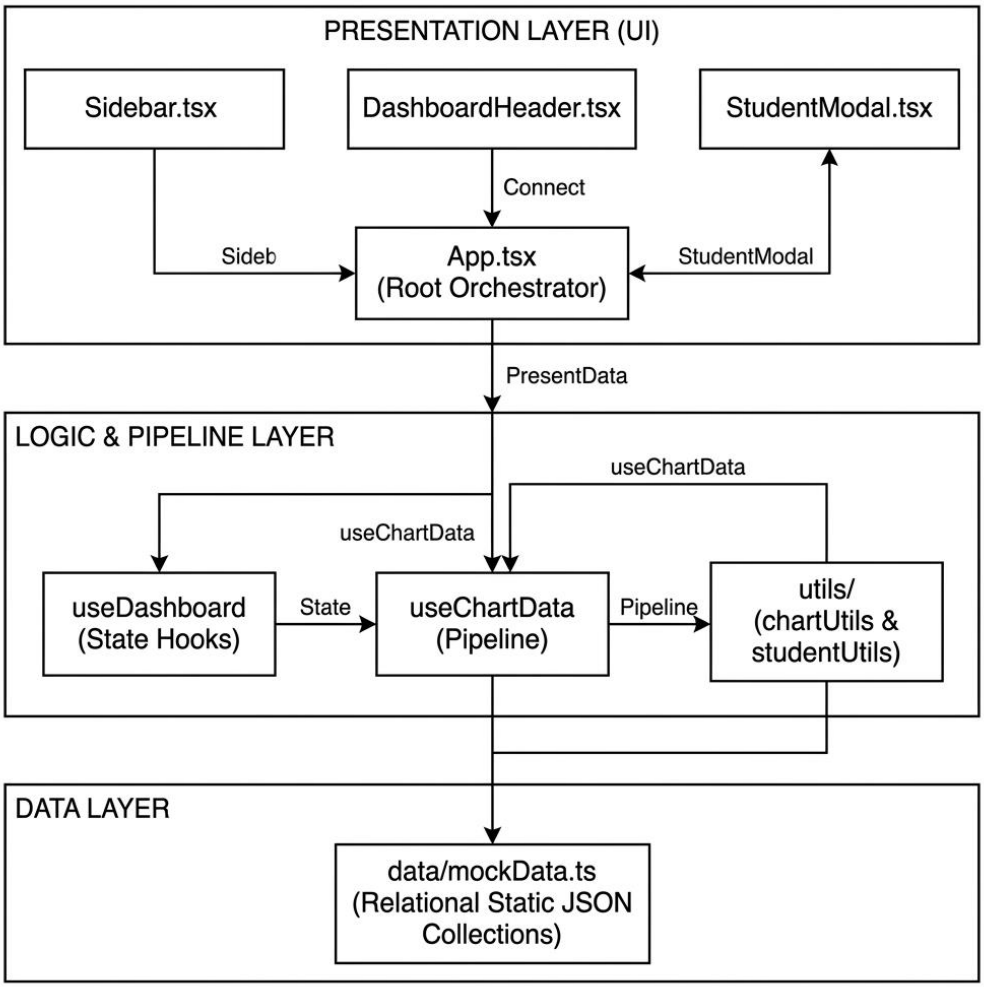


***Figure C.2**:* More detailed UML Component Diagram of the dashboard.

Figure C.2 presents the UML component architecture of the dashboard, structured into three interdependent layers. The *Presentation Layer* (UI) manages user interactions and visual output, coordinating dashboard views (OverviewCharts, StudentCharts) through the App.tsx root orchestrator. The *Logic and Pipeline Layer* isolates reactive hooks (useDashboard for state handling and useChartData for data filtering) from direct DOM rendering, relying on pure calculation helpers within the utils/ directory to maintain functional separation. Finally, the *Data Layer* acts as a static data warehouse defined in data/mockData.ts, supplying baseline scores, submission histories, and student metrics to the upper layers. Together, these components illustrate a clean modular design that supports maintainability, scalability, and clear data flow across the system.

D Appendix: Pre-questionnaire Graphs

Here the charts and graphics obtained from the pre-questionnaire can be found. Our conclusions and analysis comes from this data and the post-questionnaire data.

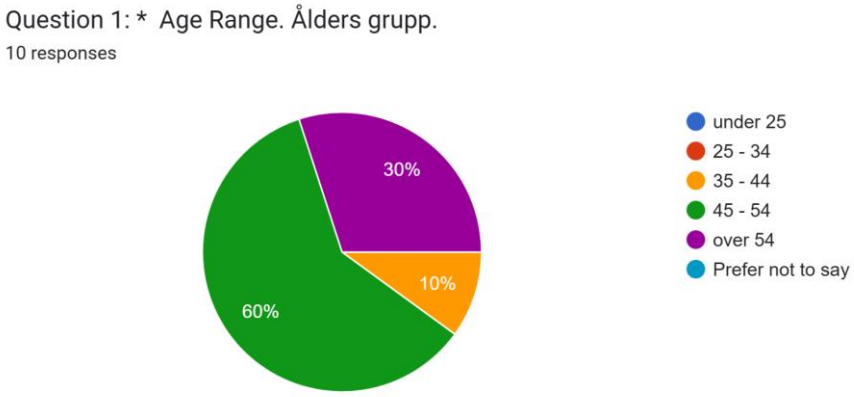


***Figure D.1**:* Participant age.

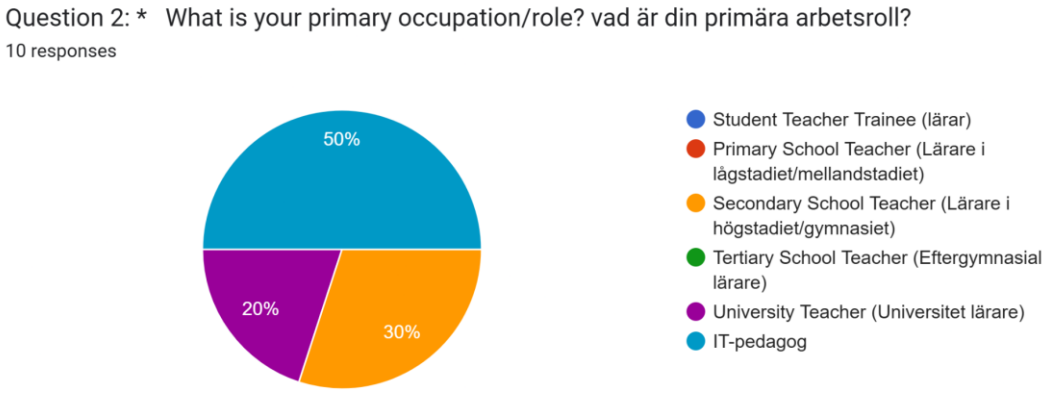


***Figure D.2**:* Participant role.

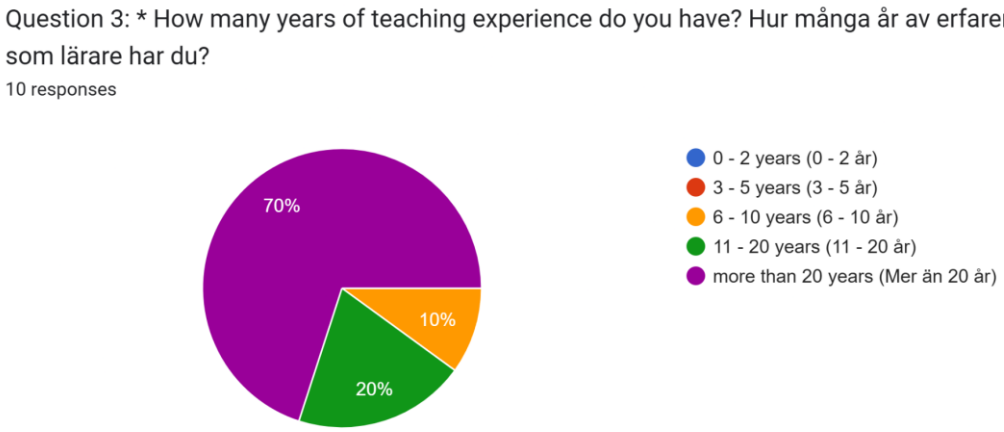


***Figure D.3**:* Participant years of experience.

Question 4: * How would you rate your skills with digital teaching tools? Hur skulle du betygsätta din färdigheter med digitala undervisnings verktyg?

10 responses

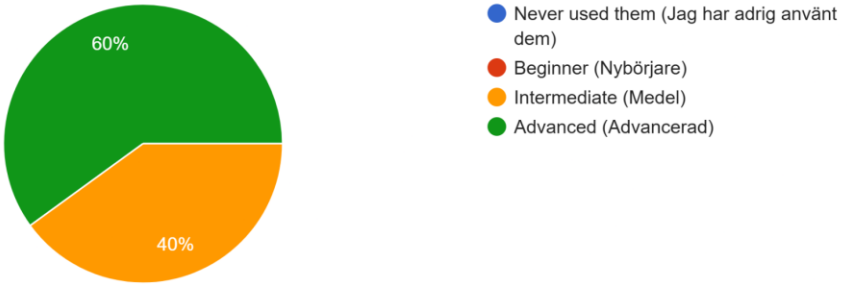


***Figure D.4**:* Participant digital teaching tools skill level.

Question 5: * Do you use digital teaching tools in class? (Använder du digitala läro verktyg i klassrummet)

10 responses

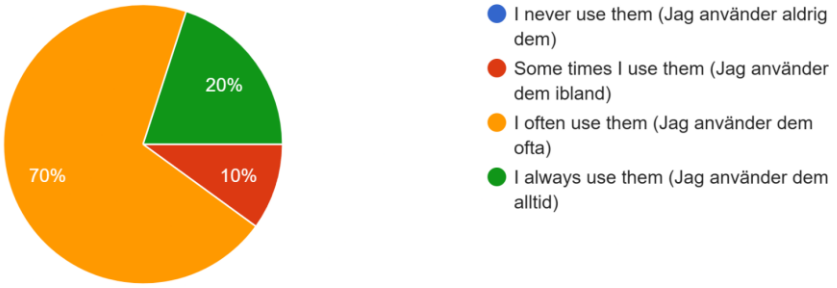


***Figure D.5**:* Participant digital teaching tool usage.

Question 6: * Have you previously used Nextgen platform? Har du tidigare använt Nextgen platformen?

10 responses

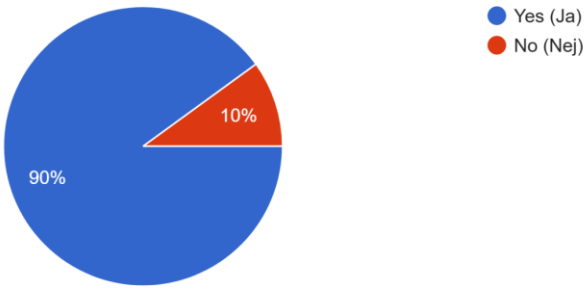


***Figure D.6**:* Participant experience with Nextgen platform.

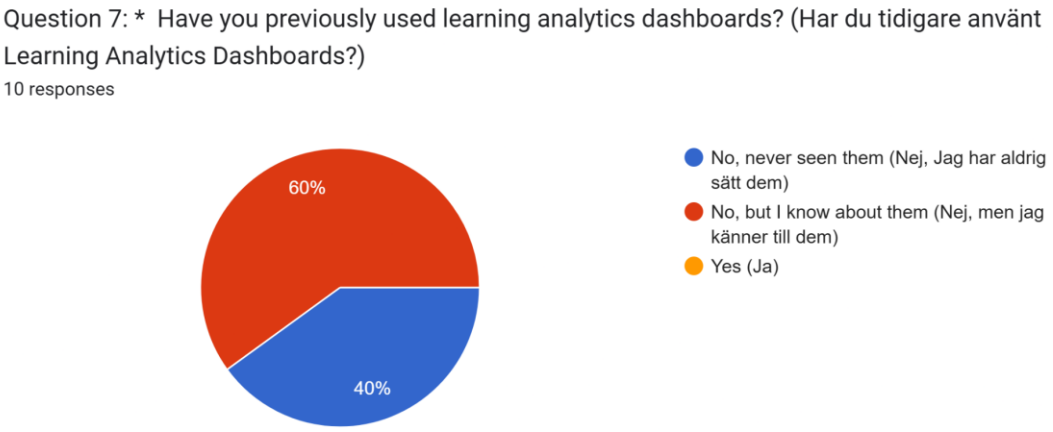


***Figure D.7**:* Participant LA dashboard experience.

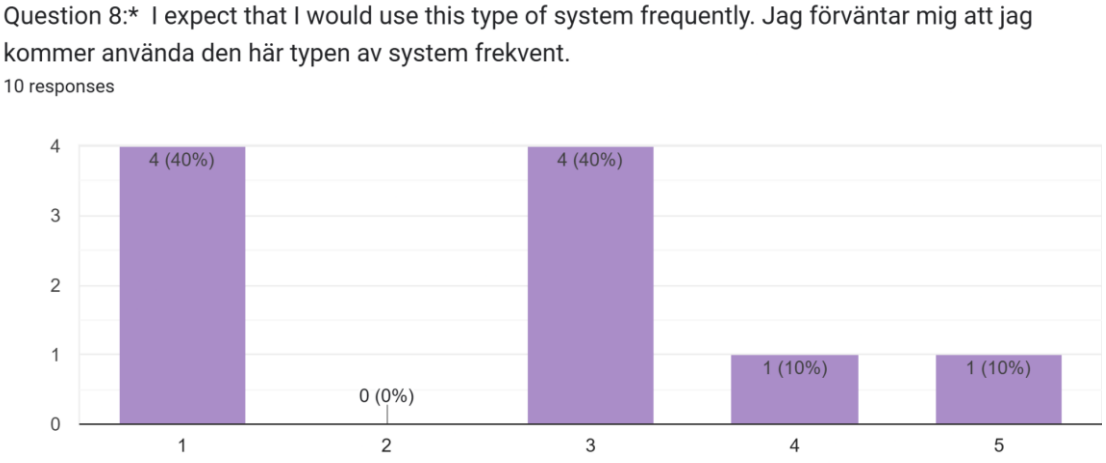


***Figure D.8**:* Pre-questionnaire question 1.

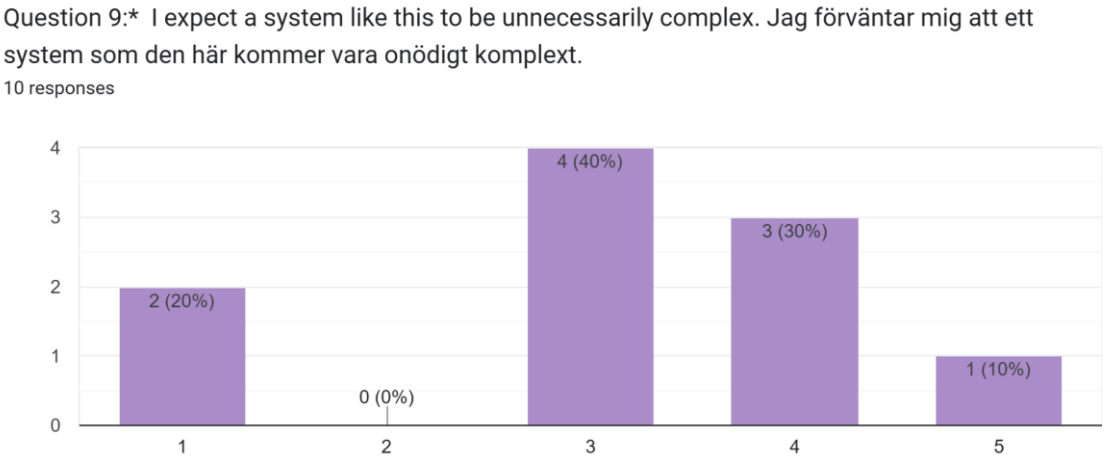


***Figure D.9**:* Pre-questionnaire question 2.

Question 10:* I expect a system like this to be easy to use. Jag förväntar mig att ett system som den här kommer vara enkelt att använda.

10 responses

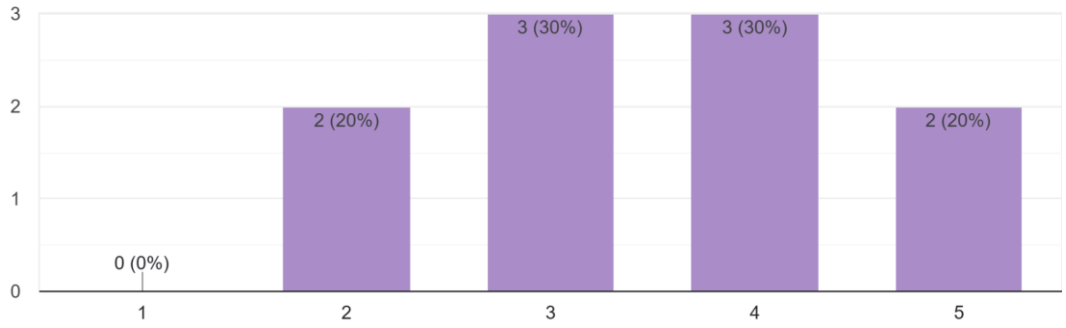


***Figure D.10**:* Pre-questionnaire question 3.

Question 11:* I think that I would need the support of a technical person to be able to use a system like this. Jag tror att jag kommer behöva hjälp av ...rson för att kunna använda ett system som den här.

10 responses

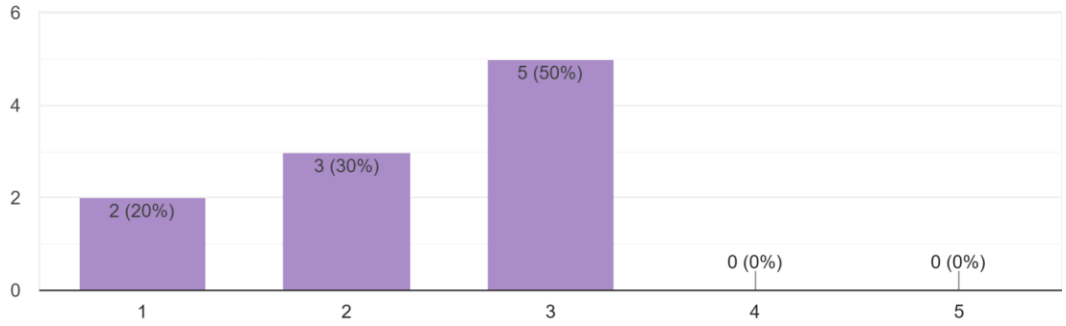


***Figure D.11**:* Pre-questionnaire question 4.

Question 12:* I expect the various functions in a system like this to be well integrated. Jag förväntar mig att de funktionerna i ett system som den här kommer vara väl integrerade.

10 responses

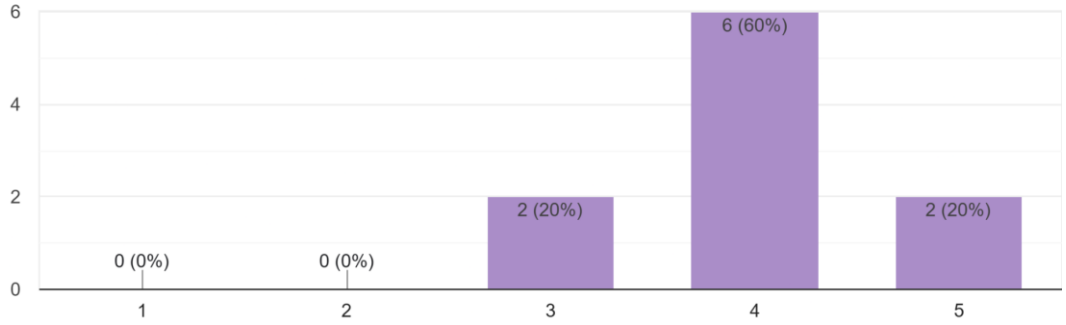


***Figure D.12**:* Pre-questionnaire question 5.

Question 13:* I expect that most people would learn to use a system like this very quickly. Jag förväntar mig att de flesta skulle lära sig att använda ett system som den här väldigt fort.

10 responses

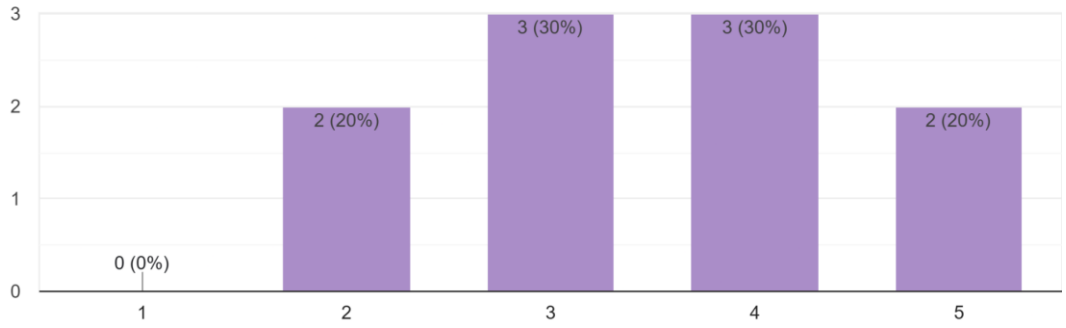


***Figure D.13**:* Pre-questionnaire question 6.

Question 14:* I expect that a system like this is very unmanageable to use. Jag förväntar mig att ett system som den här är väldigt svårhanterligt att använda.

10 responses

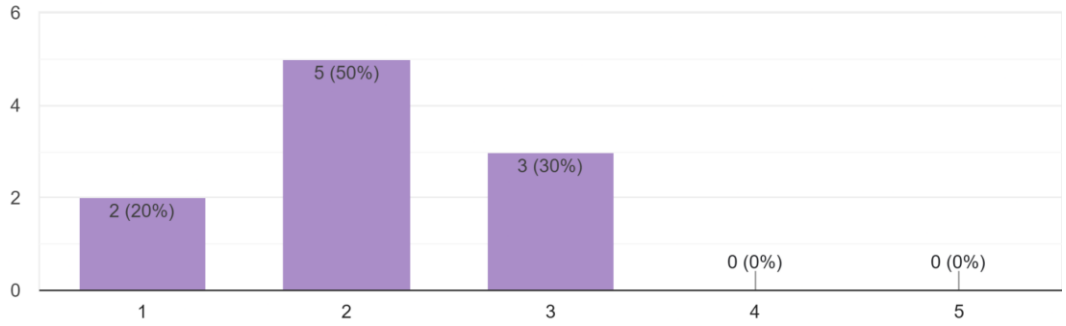


***Figure D.14**:* Pre-questionnaire question 7.

Question 15:* I expect to feel very confident using a system like this. Jag förväntar mig att känna mig säker på att använda ett system som den här.

10 responses

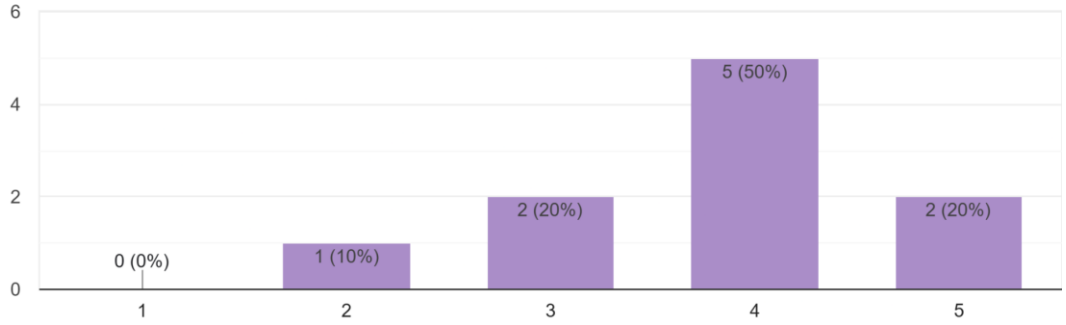


***Figure D.15**:* Pre-questionnaire question 8.

Question 16:* I expect that i need to learn a lot of things before I could get going with a system like this. Jag förväntar mig att behöva lära mig mycket...nan jag kan börja använda ett system som den här.
10 responses

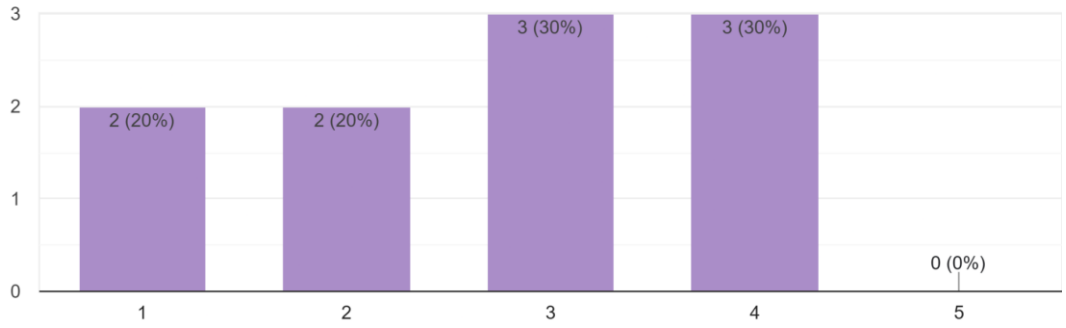


***Figure D.16**:* Pre-questionnaire question 9.

Question 17:* I expect that using a Learning Analytics Dashboard would improve my job performance. Jag förväntar mig att användning av e... Dashboard skulle förbättra min arbetsprestation.
10 responses

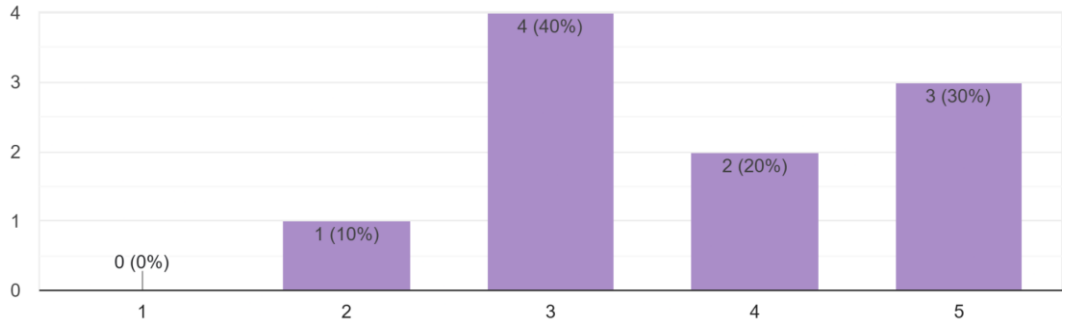


***Figure D.17**:* Pre-questionnaire question 10.

Question 18:* I expect that a Learning Analytics Dashboard would be useful in my job. Jag förväntar mig att ett Learning Analytics Dashboard skulle vara användbart för i mitt arbete.
10 responses

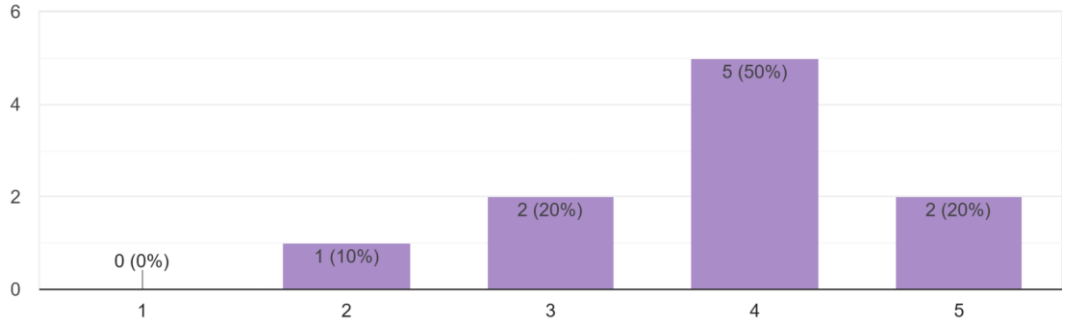


***Figure D.18**:* Pre-questionnaire question 11.

Question 19:* I expect that i will be able to correctly interpret the visualisations shown in a Learning Analytics Dashboard. Jag förväntar mig att jag kom...ngarna i ett Learning Analytics Dashboard korrekt.
10 responses

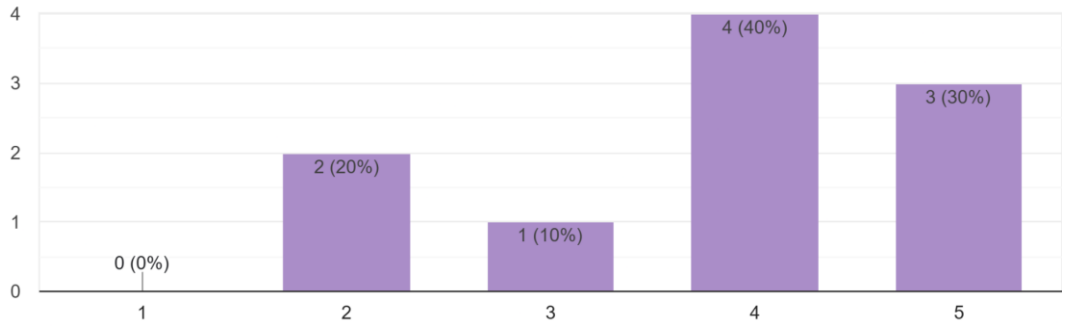


***Figure D.19**:* Pre-questionnaire question 12.

Question 20:* I expect that i will be able to translate insights from a Learning Analytics Dashboard into concrete actions in the classroom. Jag förvänt...cs Dashboard till konkreta åtgärder i klassurmmet.
10 responses

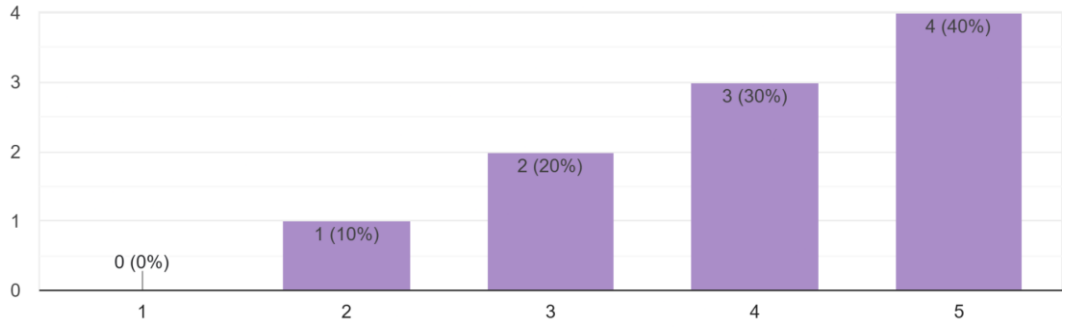


***Figure D.20**:* Pre-questionnaire question 13.

Question 21:* I expect that i will be able to use data to identify students who need additional support. Jag förväntar mig att kunna använda data för att identifiera studenter som behöver extra stöd.
10 responses

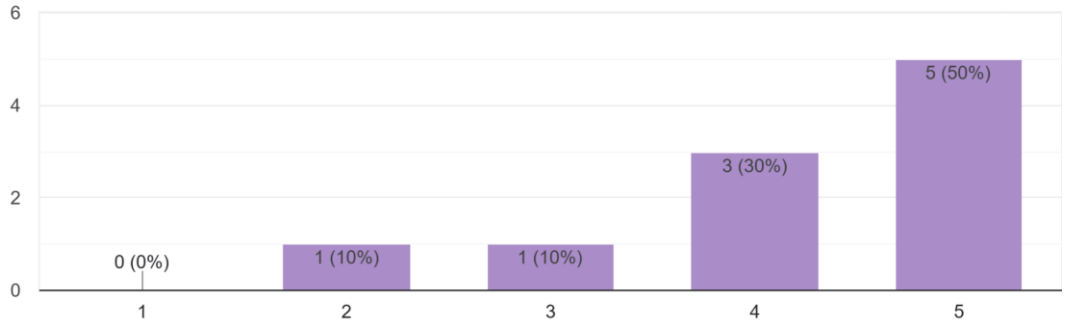


***Figure D.21**:* Pre-questionnaire question 14.

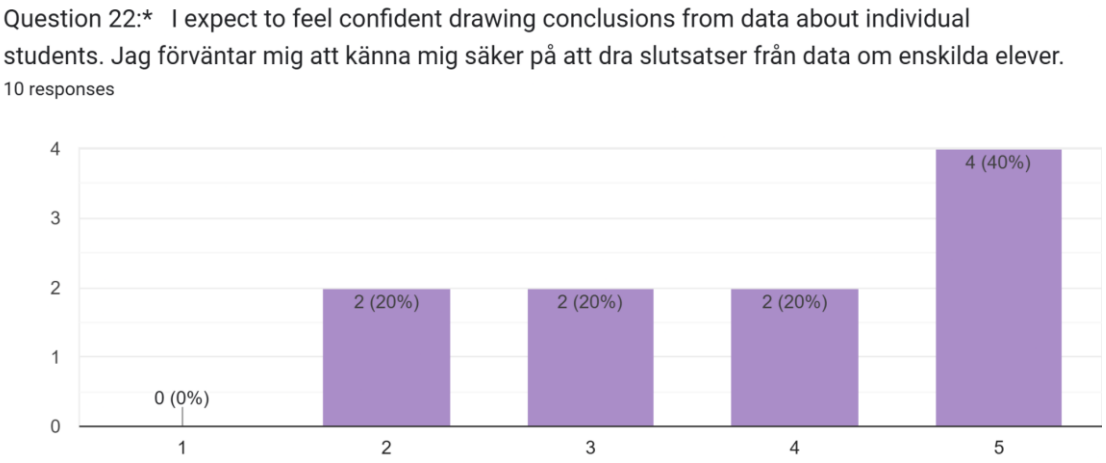


***Figure D.22**:* Pre-questionnaire question 15.

E Appendix: Post-questionnaire Graphs

Here the charts and graphics obtained from the post-questionnaire can be found.

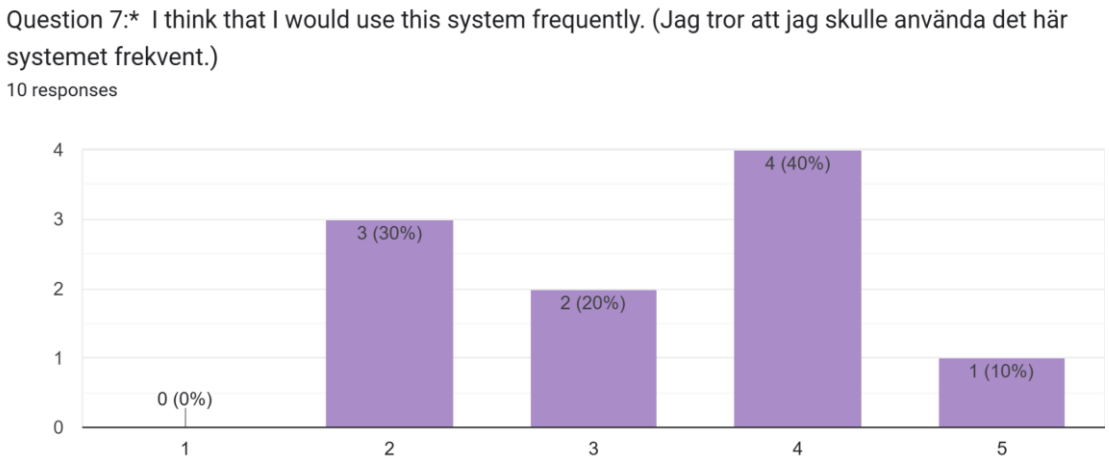


***Figure E.1**:* Post-questionnaire question 1.

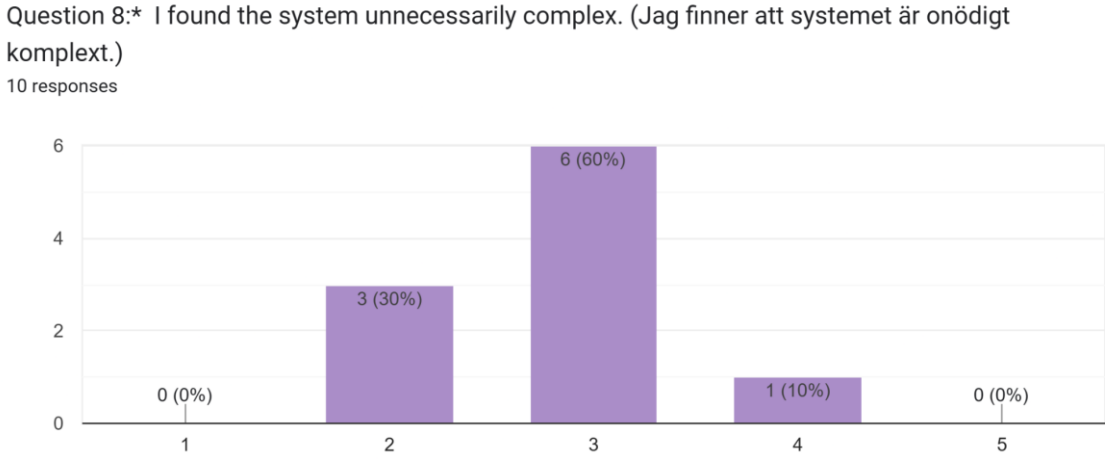


***Figure E.2**:* Post-questionnaire question 2.

Question 9:* I thought the system was easy to use. (Jag tyckte att systemet var enkelt att använda.)

10 responses

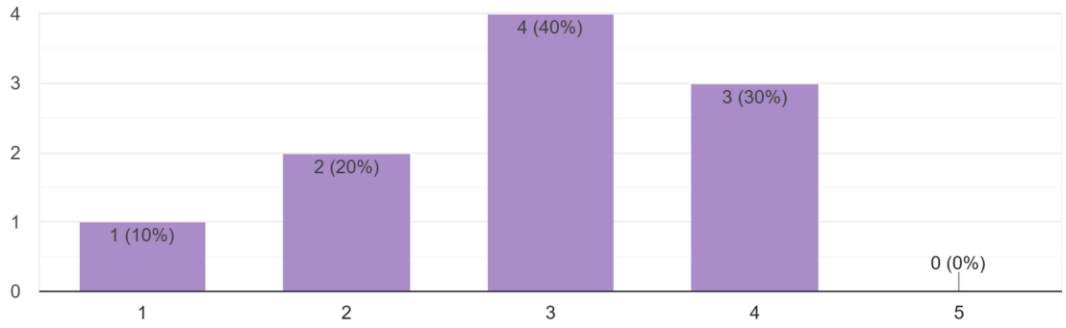


***Figure E.3**:* Post-questionnaire question 3.

Question 10:* I think that I would need the support of a technical person to be able to use this system. (Jag tror att jag kommer behöva hjälp av e...knisk person för att kunna använda detta system.)

10 responses

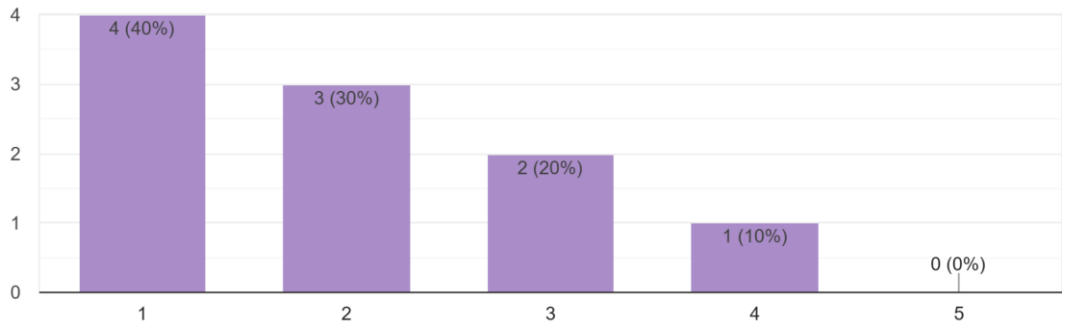


***Figure E.4**:* Post-questionnaire question 4.

Question 11:* I found the various functions in this system were well integrated. (Jag tycker att de funktionerna i systemet var väl integrerade.)

10 responses

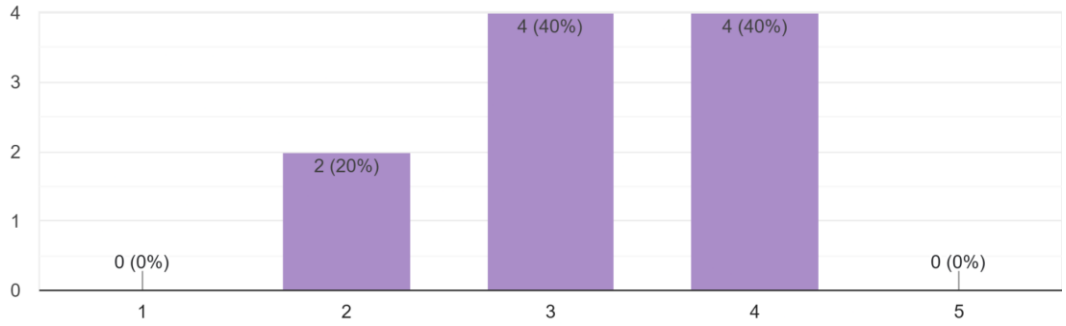


***Figure E.5**:* Post-questionnaire question 5.

Question 12:* I would imagine that most people would learn to use this system very quickly. (Jag tror att de flesta skulle lära sig att använda detta system väldigt fort.)

10 responses

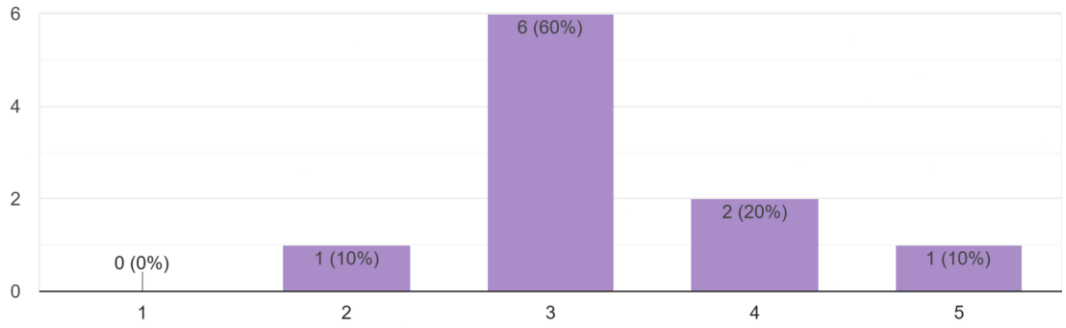


***Figure E.6**:* Post-questionnaire question 6.

Question 13:* I found the system very unmanageable to use. (Jag tycker att systemet var väldigt svårhanterligt att använda.)

10 responses

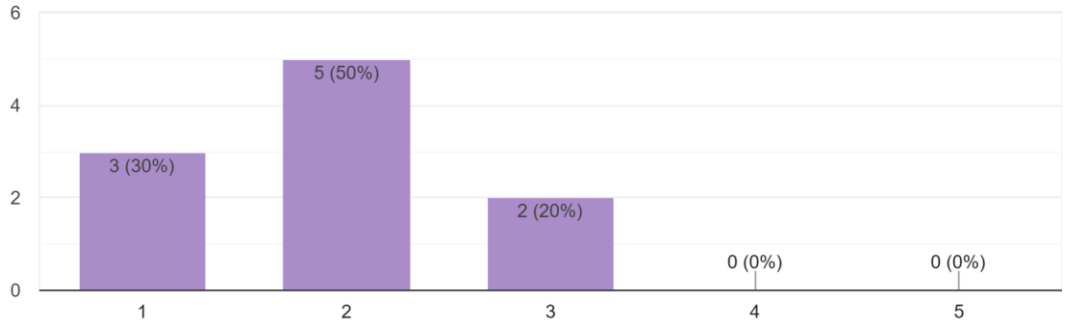


***Figure E.7**:* Post-questionnaire question 7.

Question 14:* I felt very confident using the system . (Jag känner mig säker på att använda systemet.)

10 responses

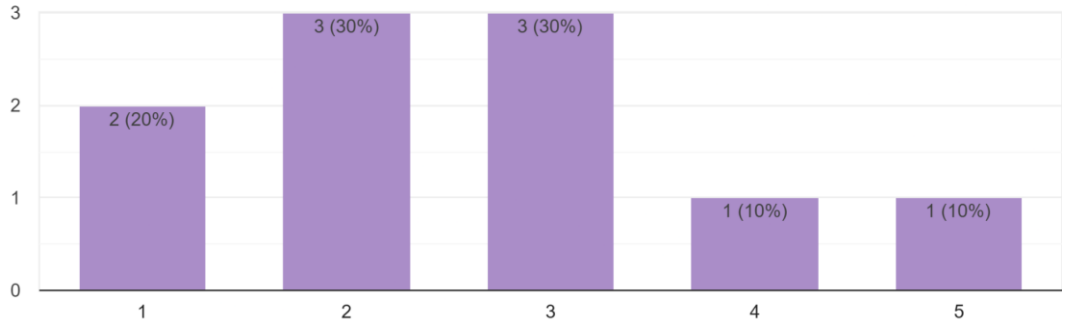


***Figure E.8**:* Post-questionnaire question 8.

Question 15:* I needed to learn a lot of things before I could get going with this system. (Jag behövde lära mig mycket innan jag kunde börja använda ett systemet.)

10 responses

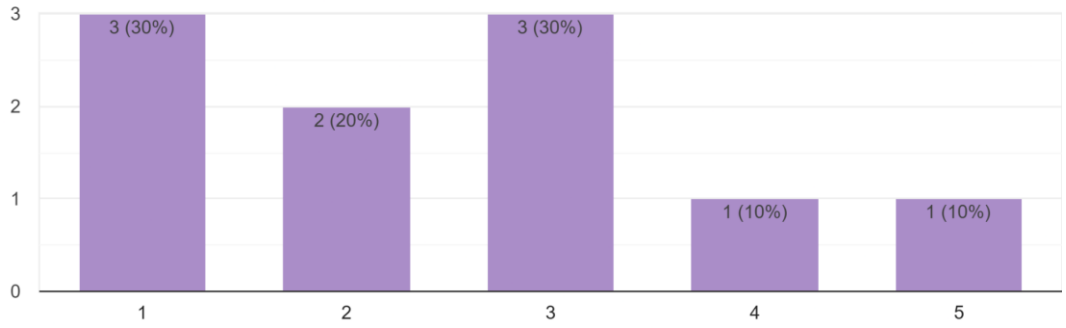


Figure E.9: Post-questionnaire question 9.

Question 16:* Using the Learning Analytics Dashboard would improve my job performance. (Användning av denna Learning Analyt...ashboard skulle förbättra min arbetsprestation.)

10 responses

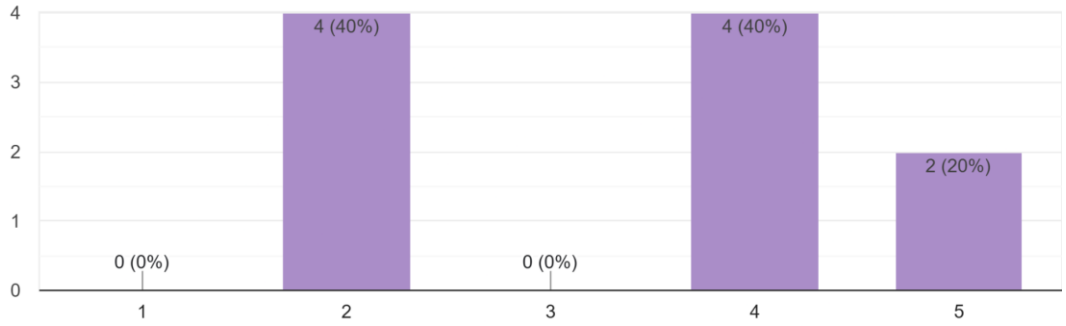


Figure E.10: Post-questionnaire question 10.

Question 17:* I would find the Learning Analytics Dashboard would be useful in my job. (Jag finner att denna Learning Analytics Dashboard skulle vara användbart för i mitt arbete.)

10 responses

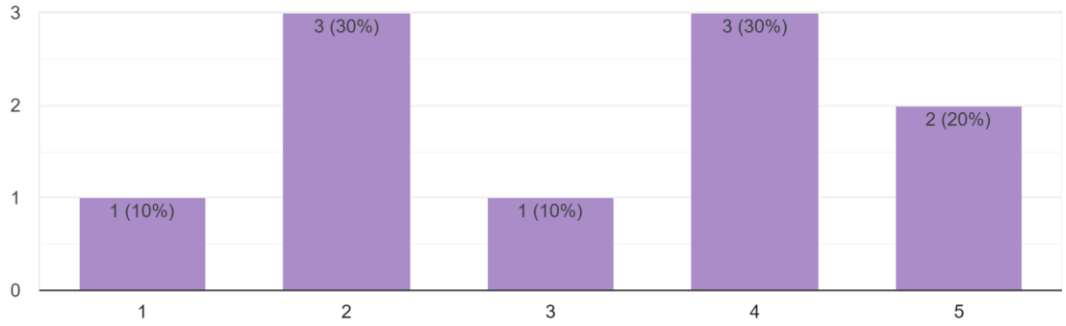


Figure E.11: Post-questionnaire question 11.

Question 18:* I can correctly interpret the visualisations shown in the Learning Analytics Dashboard. (Jag kan tolka visualiseringarna i denna Learning Analytics Dashboard korrekt.)
10 responses

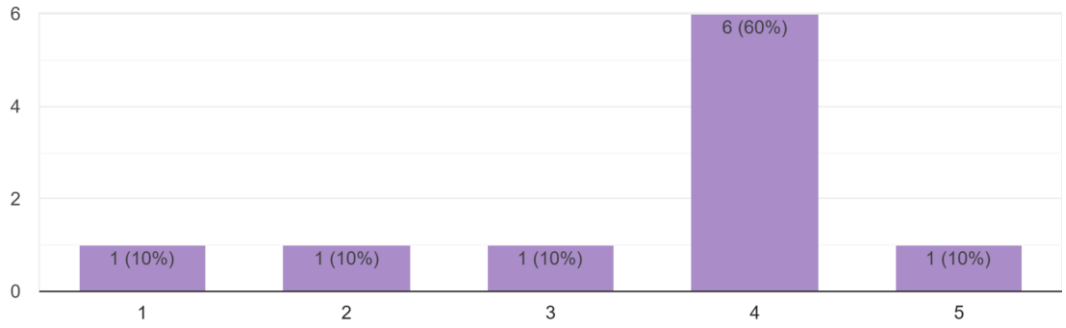


***Figure E.12**:* Post-questionnaire question 12.

Question 19:* I translate insights from the Learning Analytics Dashboard into concrete actions in the classroom. (Jag översätter insikter från denna ...s Dashboard till konkreta åtgärder i klassurmmet.)
10 responses

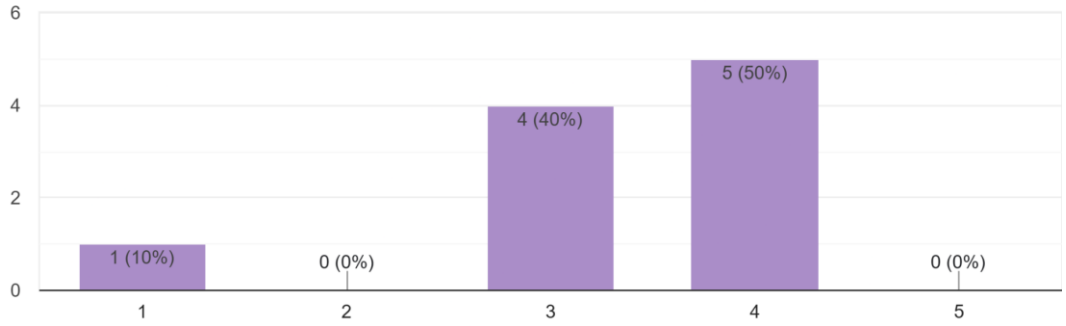


***Figure E.13**:* Post-questionnaire question 13.

Question 20:* I use data to identify students who need additional support. (Jag använder data för att identifiera studenter som behöver extra stöd.)
10 responses

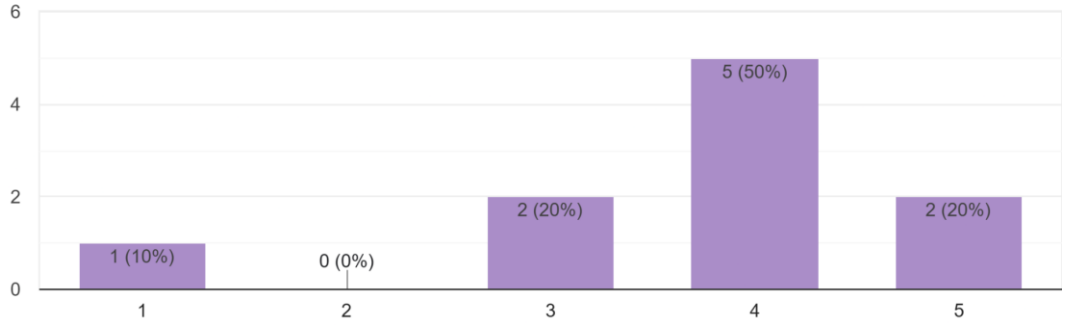


***Figure E.14**:* Post-questionnaire question 14.

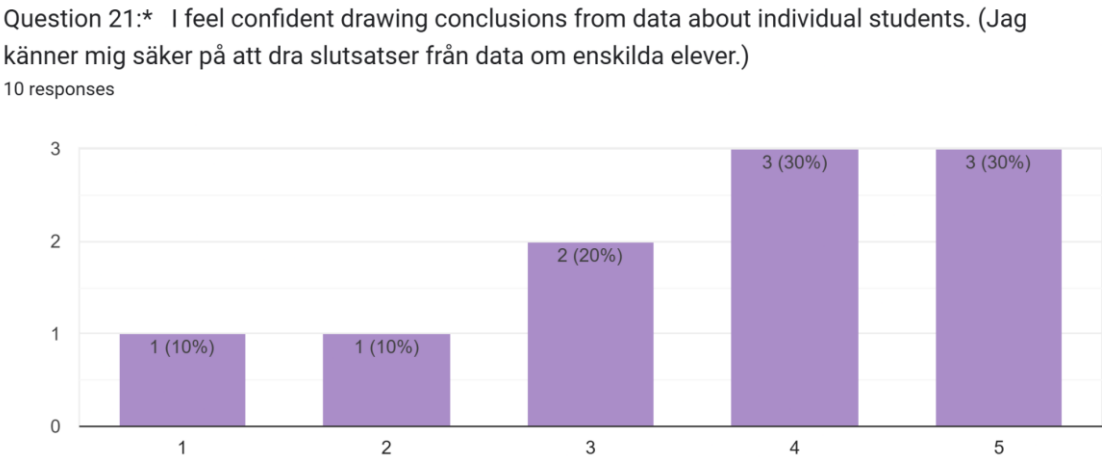


***Figure E.15**:* Post-questionnaire question 15.

F Appendix: Workshop tasks

Identify which students aren't active
Identify 3 students that are struggling at the class
Identify 3 students that excel at the class
Check how much time did Evelin spend in the app during last week on tuesday
Check what you think are the needs of the struggling students you previously identified
What are the overall needs of the class?
Do you think all of those 3 students struggling put in the work or not?

Table F.1:.